

Graduate Texts in Mathematics

Kenneth S. Brown

Cohomology of Groups

群的上同调

Springer

世界图书出版公司
www.wpcbj.com.cn

Graduate Texts in Mathematics 87

Editorial Board

S. Axler F.W. Gehring P.R. Halmos

Springer

New York

Berlin

Heidelberg

Barcelona

Budapest

Hong Kong

London

Milan

Paris

Santa Clara

Singapore

Tokyo

Graduate Texts in Mathematics

- 1 TAKEUTI/ZARING. Introduction to Axiomatic Set Theory. 2nd ed.
- 2 OXToby. Measure and Category. 2nd ed.
- 3 SCHAEFER. Topological Vector Spaces.
- 4 HILTON/STAMMBACH. A Course in Homological Algebra.
- 5 MAC LANE. Categories for the Working Mathematician.
- 6 HUGHES/PIPER. Projective Planes.
- 7 SERRE. A Course in Arithmetic.
- 8 TAKEUTI/ZARING. Axiomatic Set Theory.
- 9 HUMPHREYS. Introduction to Lie Algebras and Representation Theory.
- 10 COHEN. A Course in Simple Homotopy Theory.
- 11 CONWAY. Functions of One Complex Variable I. 2nd ed.
- 12 BEALS. Advanced Mathematical Analysis.
- 13 ANDERSON/FULLER. Rings and Categories of Modules. 2nd ed.
- 14 GOLUBITSKY/GUILLEMIN. Stable Mappings and Their Singularities.
- 15 BERBERIAN. Lectures in Functional Analysis and Operator Theory.
- 16 WINTER. The Structure of Fields.
- 17 ROSENBLATT. Random Processes. 2nd ed.
- 18 HALMOS. Measure Theory.
- 19 HALMOS. A Hilbert Space Problem Book. 2nd ed.
- 20 HUSEMOLLER. Fibre Bundles. 3rd ed.
- 21 HUMPHREYS. Linear Algebraic Groups.
- 22 BARNES/MACK. An Algebraic Introduction to Mathematical Logic.
- 23 GREUB. Linear Algebra. 4th ed.
- 24 HOLMES. Geometric Functional Analysis and Its Applications.
- 25 HEWITT/STROMBERG. Real and Abstract Analysis.
- 26 MANES. Algebraic Theories.
- 27 KELLEY. General Topology.
- 28 ZARISKI/SAMUEL. Commutative Algebra. Vol. I.
- 29 ZARISKI/SAMUEL. Commutative Algebra. Vol. II.
- 30 JACOBSON. Lectures in Abstract Algebra I. Basic Concepts.
- 31 JACOBSON. Lectures in Abstract Algebra II. Linear Algebra.
- 32 JACOBSON. Lectures in Abstract Algebra III. Theory of Fields and Galois Theory.
- 33 HIRSCH. Differential Topology.
- 34 SPITZER. Principles of Random Walk. 2nd ed.
- 35 WERMER. Banach Algebras and Several Complex Variables. 2nd ed.
- 36 KELLEY/NAMIOKA et al. Linear Topological Spaces.
- 37 MONK. Mathematical Logic.
- 38 GRAUBERT/FRITZSCHE. Several Complex Variables.
- 39 ARVESON. An Invitation to C^* -Algebras.
- 40 KEMENY/SNELL/KNAPP. Denumerable Markov Chains. 2nd ed.
- 41 APOSTOL. Modular Functions and Dirichlet Series in Number Theory. 2nd ed.
- 42 SERRE. Linear Representations of Finite Groups.
- 43 GILLMAN/JERISON. Rings of Continuous Functions.
- 44 KENDIG. Elementary Algebraic Geometry.
- 45 LOÈVE. Probability Theory I. 4th ed.
- 46 LOÈVE. Probability Theory II. 4th ed.
- 47 MOISE. Geometric Topology in Dimensions 2 and 3.
- 48 SACHS/WU. General Relativity for Mathematicians.
- 49 GRUENBERG/WEIR. Linear Geometry. 2nd ed.
- 50 EDWARDS. Fermat's Last Theorem.
- 51 KLINGENBERG. A Course in Differential Geometry.
- 52 HARTSHORNE. Algebraic Geometry.
- 53 MANIN. A Course in Mathematical Logic.
- 54 GRAVER/WATKINS. Combinatorics with Emphasis on the Theory of Graphs.
- 55 BROWN/PEARCY. Introduction to Operator Theory I: Elements of Functional Analysis.
- 56 MASSEY. Algebraic Topology: An Introduction.
- 57 CROWELL/FOX. Introduction to Knot Theory.
- 58 KOBLITZ. p -adic Numbers, p -adic Analysis, and Zeta-Functions. 2nd ed.
- 59 LANG. Cyclotomic Fields.
- 60 ARNOLD. Mathematical Methods in Classical Mechanics. 2nd ed.

continued after index

Kenneth S. Brown

Cohomology of Groups



Springer

Kenneth S. Brown
Department of Mathematics
White Hall
Cornell University
Ithaca, NY 14853
U.S.A.

Editorial Board

S. Axler
Department of
Mathematics
Michigan State University
East Lansing, MI 48824
USA

F.W. Gehring
Department of
Mathematics
University of Michigan
Ann Arbor, MI 48109
USA

P.R. Halmos
Department of
Mathematics
Santa Clara University
Santa Clara, CA 95053
USA

Library of Congress Cataloging in Publication Data

Brown, Kenneth S.

Cohomology of groups.

(Graduate texts in mathematics; 87)

Bibliography: p.

Includes index.

1. Groups, Theory of. 2. Homology theory.

I. Title. II. Series.

QA171.B876 512'.22 82-733
AACR2

© 1982 by Springer-Verlag New York Inc.

All rights reserved. No part of this book may be translated or reproduced in any form without written permission from Springer-Verlag, 175 Fifth Avenue, New York, New York 10010, U.S.A.

This reprint has been authorized by Springer-Verlag (Berlin/Heidelberg/New York) for sale in the Mainland China only and not for export therefrom.

ISBN 978-1-4684-9329-0 ISBN 978-1-4684-9327-6 (eBook)
DOI 10.1007/978-1-4684-9327-6

Preface

This book is based on a course given at Cornell University and intended primarily for second-year graduate students. The purpose of the course was to introduce students who knew a little algebra and topology to a subject in which there is a very rich interplay between the two. Thus I take neither a purely algebraic nor a purely topological approach, but rather I use both algebraic and topological techniques as they seem appropriate.

The first six chapters contain what I consider to be the basics of the subject. The remaining four chapters are somewhat more specialized and reflect my own research interests. For the most part, the only prerequisites for reading the book are the elements of algebra (groups, rings, and modules, including tensor products over non-commutative rings) and the elements of algebraic topology (fundamental group, covering spaces, simplicial and CW-complexes, and homology). There are, however, a few theorems, especially in the later chapters, whose proofs use slightly more topology (such as the Hurewicz theorem or Poincaré duality). The reader who does not have the required background in topology can simply take these theorems on faith.

There are a number of exercises, some of which contain results which are referred to in the text. A few of the exercises are marked with an asterisk to warn the reader that they are more difficult than the others or that they require more background.

I am very grateful to R. Bieri, J-P. Serre, U. Stambach, R. Strebel, and C. T. C. Wall for helpful comments on a preliminary version of this book.

Notational Conventions

All rings (including graded rings) are assumed to be associative and to have an identity. The latter is required to be preserved by ring homomorphisms. Modules are understood to be *left* modules, unless the contrary is explicitly stated. Similarly, group actions are generally understood to be left actions.

If a group G acts on a set X , I will usually write X/G instead of $G \backslash X$ for the orbit set, even if G is acting on the left. One exception to this concerns the notation for the set of cosets of a subgroup H in a group G . Here we are talking about the left or right translation action of H on G , and I will always be careful to put the H on the appropriate side. Thus $G/H = \{gH: g \in G\}$ and $H \backslash G = \{Hg: g \in G\}$.

A symbol such as

$$\sum_{g \in G/H} f(g)$$

indicates that f is a function on G such that $f(g)$ depends only on the coset gH of g ; the sum is then taken over a set of coset representatives.

Finally, I use the “topologists’ notation”

$$\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z};$$

in particular, $\mathbb{Z}_p$ denotes the integers mod p , not the p -adic integers.

Contents

Introduction	1
CHAPTER I	
Some Homological Algebra	4
0. Review of Chain Complexes	4
1. Free Resolutions	10
2. Group Rings	12
3. G -Modules	13
4. Resolutions of $\mathbb{Z}$ Over $\mathbb{Z}G$ via Topology	14
5. The Standard Resolution	18
6. Periodic Resolutions via Free Actions on Spheres	20
7. Uniqueness of Resolutions	21
8. Projective Modules	26
Appendix. Review of Regular Coverings	31
CHAPTER II	
The Homology of a Group	33
1. Generalities	33
2. Co-invariants	34
3. The Definition of H_*G	35
4. Topological Interpretation	36
5. Hopf's Theorems	41
6. Functoriality	48
7. The Homology of Amalgamated Free Products	49
Appendix. Trees and Amalgamations	52
CHAPTER III	
Homology and Cohomology with Coefficients	55
0. Preliminaries on $\otimes_G$ and Hom_G	55
1. Definition of $H_*(G, M)$ and $H^*(G, M)$	56

2. Tor and Ext	60
3. Extension and Co-extension of Scalars	62
4. Injective Modules	65
5. Induced and Co-induced Modules	67
6. H_* and H^* as Functors of the Coefficient Module	71
7. Dimension Shifting	74
8. H_* and H^* as Functors of Two Variables	78
9. The Transfer Map	80
10. Applications of the Transfer	83

CHAPTER IV

Low Dimensional Cohomology and Group Extensions 86

1. Introduction	86
2. Split Extensions	87
3. The Classification of Extensions with Abelian Kernel	91
4. Application: p -Groups with a Cyclic Subgroup of Index p	97
5. Crossed Modules and H^3 (Sketch)	102
6. Extensions With Non-Abelian Kernel (Sketch)	104

CHAPTER V

Products 107

1. The Tensor Product of Resolutions	107
2. Cross-products	108
3. Cup and Cap Products	109
4. Composition Products	114
5. The Pontryagin Product	117
6. Application: Calculation of the Homology of an Abelian Group	121

CHAPTER VI

Cohomology Theory of Finite Groups 128

1. Introduction	128
2. Relative Homological Algebra	129
3. Complete Resolutions	131
4. Definition of $\hat{H}^*$	134
5. Properties of $\hat{H}^*$	136
6. Composition Products	142
7. A Duality Theorem	144
8. Cohomologically Trivial Modules	148
9. Groups with Periodic Cohomology	153

CHAPTER VII

Equivariant Homology and Spectral Sequences 161

1. Introduction	161
2. The Spectral Sequence of a Filtered Complex	161

3. Double Complexes	164
4. Example: The Homology of a Union	166
5. Homology of a Group with Coefficients in a Chain Complex	168
6. Example: The Hochschild–Serre Spectral Sequence	171
7. Equivariant Homology	172
8. Computation of d^1	175
9. Example: Amalgamations	178
10. Equivariant Tate Cohomology	180

CHAPTER VIII	
Finiteness Conditions	183
1. Introduction	183
2. Cohomological Dimension	184
3. Serre's Theorem	190
4. Resolutions of Finite Type	191
5. Groups of Type FP_n	197
6. Groups of Type FP and FL	199
7. Topological Interpretation	205
8. Further Topological Results	210
9. Further Examples	213
10. Duality Groups	219
11. Virtual Notions	225

CHAPTER IX	
Euler Characteristics	230
1. Ranks of Projective Modules: Introduction	230
2. The Hattori–Stallings Rank	231
3. Ranks Over Commutative Rings	235
4. Ranks Over Group Rings; Swan's Theorem	239
5. Consequences of Swan's Theorem	242
6. Euler Characteristics of Groups: The Torsion-Free Case	246
7. Extension to Groups with Torsion	249
8. Euler Characteristics and Number Theory	253
9. Integrality Properties of $\chi(\Gamma)$	257
10. Proof of Theorem 9.3; Finite Group Actions	258
11. The Fractional Part of $\chi(\Gamma)$	261
12. Acyclic Covers; Proof of Lemma 11.2	265
13. The p -Fractional Part of $\chi(\Gamma)$	266
14. A Formula for $\chi_\Gamma(\mathcal{A})$	270

CHAPTER X	
Farrell Cohomology Theory	273
1. Introduction	273
2. Complete Resolutions	273
3. Definition and Properties of $\hat{H}^*(\Gamma)$	277

4. Equivariant Farrell Cohomology	281
5. Cohomologically Trivial Modules	287
6. Groups with Periodic Cohomology	288
7. $\hat{H}^*(\Gamma)$ and the Ordered Set of Finite Subgroups of Γ	291
 References	 295
 Notation Index	 301
 Index	 303

Introduction

The cohomology theory of groups arose from both topological and algebraic sources. The starting point for the topological aspect of the theory was the work of Hurewicz [1936] on “aspherical spaces.” About a year earlier, Hurewicz had introduced the higher homotopy groups $\pi_n X$ of a space X ($n \geq 2$). He now singled out for study those path-connected spaces X whose higher homotopy groups are all trivial, but whose fundamental group $\pi = \pi_1 X$ need not be trivial. Such spaces are called *aspherical*.

Hurewicz proved, among other things, that the homotopy type of an aspherical space X is completely determined by its fundamental group π . In particular, the homology groups of X depend only on π ; it is therefore reasonable to think of them as *homology groups of π* . [This terminology, however, was not introduced until the 1940’s.] Throughout the remainder of this introduction, then, we will write $H_* \pi$ for the homology of any aspherical space with fundamental group π . (Similarly, we could define homology and cohomology groups of π with arbitrary coefficients.) As a simple example, note that $H_2(\mathbb{Z} \oplus \mathbb{Z}) = \mathbb{Z}$. [Take X to be the torus.] Although Hurewicz considered only the *uniqueness* and not the *existence* of aspherical spaces, there does in fact exist an aspherical space with any given group as fundamental group. Thus our topological definition of group homology applies to all groups

For any group π we obviously have $H_0 \pi = \mathbb{Z}$ and $H_1 \pi = \pi_{ab}$, the latter being the abelianization of π , i.e., π modulo its commutator subgroup. For $n \geq 2$, however, it is by no means clear how to describe $H_n \pi$ algebraically. The first progress in this direction was made by Hopf [1942], who expressed $H_2 \pi$ in purely algebraic terms, and who gave further evidence of its importance in topology by proving the following theorem: for any path-connected space X with fundamental group π , there is an exact sequence

$$(0.1) \quad \pi_2 X \rightarrow H_2 X \rightarrow H_2 \pi \rightarrow 0.$$

[To put this result in perspective, one should recall that Hurewicz had introduced homomorphisms $h_n: \pi_n X \rightarrow H_n X$ ($n \geq 2$) and had shown that h_n is an isomorphism if $\pi_i X = 0$ for $i < n$. In particular, h_2 is an isomorphism if $\pi = \pi_1 X = 0$. When π is non-trivial, however, h_2 is in general neither injective nor surjective, and Hopf's theorem gives a precise description, in terms of π , of the extent to which surjectivity fails.]

Hopf's description of $H_2 \pi$, incidentally, went as follows: Choose a presentation of π as F/R , where F is free and $R \triangleleft F$; then

$$(0.2) \quad H_2 \pi = R \cap [F, F]/[R, F],$$

where $[A, B]$ for $A, B \subseteq F$ denotes the subgroup generated by the commutators $[a, b] = aba^{-1}b^{-1}$ ($a \in A, b \in B$).

Following Hopf's paper there was a rapid development of the subject by Eckmann, Eilenberg–MacLane, Freudenthal, and Hopf. (See MacLane [1978] for some comments about this development.) In particular, one had by the mid-1940's a purely algebraic definition of group homology and cohomology, from which it became clear that the subject was of interest to algebraists as well as topologists. Indeed, the low-dimensional cohomology groups were seen to coincide with groups which had been introduced much earlier in connection with various algebraic problems. H^1 , for instance, consists of equivalence classes of "crossed homomorphisms" or "derivations." And H^2 consists of equivalence classes of "factor sets," the study of which goes back to Schur [1904], Schreier [1926], and Brauer [1926]. Even H^3 had appeared in an algebraic context (Teichmüller [1940]). These are the algebraic sources of group cohomology referred to at the beginning of this introduction. (Of course, there had been nothing in this algebraic work to suggest that there was an underlying "homology theory"; this had to wait for the impetus from topology.)

It is not surprising, in view of this history, that the subject of group cohomology offers possibilities for a great deal of interaction between algebra and topology. For instance a "transfer map," motivated by a classical group-theoretic construction due to Schur [1902], was introduced into group cohomology (Eckmann [1953], Artin–Tate [unpublished]) and from there into topology, where it has become an important tool. Another example is the theory of Euler characteristics of groups. This theory was motivated by topology, but it has applications to group theory and number theory.

Our approach to the subject will be as follows: We begin in Chapters I and II by defining $H_* \pi$ from the point of view of "homological algebra." This is the point of view which had evolved by the end of the 1940's. The topological motivation, however, will always be kept in sight, and we will immediately obtain the topological interpretation of $H_* \pi$ in terms of aspherical spaces. In particular, we will prove 0.1 and 0.2.

Chapter III contains more homological algebra, involving homology and cohomology with coefficients. These arise naturally in applications, both in algebra and topology. They are also an important technical tool, since they

make it possible to prove theorems by “dimension-shifting.” In Chapter IV we treat the theory of group extensions, which involves the crossed homomorphisms and factor sets mentioned above.

Chapter V introduces cup and cap products (motivated by topology), and these are then used in Chapter VI in the study of the cohomology of finite groups. Much of the material in Chapter VI (such as the “Tate cohomology theory”) was originally motivated by algebra (class field theory), but it turns out to be related to topological questions as well, such as the study of groups acting freely on spheres.

In Chapter VII we introduce spectral sequence techniques, which are used extensively in the remaining chapters. The reader is not expected to have previously seen spectral sequences; I give a reasonably self-contained treatment, omitting only some routine (but tedious) verifications.

Beginning with Chapter VIII the emphasis is on infinite groups, with the most interesting examples being groups of integral matrices. In Chapter VIII we discuss various finiteness conditions which can be imposed on such a group to guarantee that the homology has nice properties. Chapter IX treats Euler characteristics, which can be defined under suitable finiteness hypotheses. This theory, as we mentioned above, has interesting connections with number theory. Finally, Chapter X develops the “Farrell cohomology theory,” which is a generalization to infinite groups of the Tate cohomology theory for finite groups.

CHAPTER I

Some Homological Algebra

0 Review of Chain Complexes

We collect here for ease of reference some terminology and results concerning chain complexes. Much of this will be well-known to anyone who has studied algebraic topology. The reader is advised to skip this section (or skim it lightly) and refer back to it as necessary. We will omit some of the proofs; these are either easy or else can be found in standard texts, such as Dold [1972], Spanier [1966], or MacLane [1963].

Let R be an arbitrary ring. By a *graded R -module* we mean a sequence $C = (C_n)_{n \in \mathbb{Z}}$ of R -modules. If $x \in C_n$, then we say x has *degree n* and we write $\deg x = n$. A *map of degree p* from a graded R -module C to a graded R -module C' is a family $f = (f_n: C_n \rightarrow C'_{n+p})_{n \in \mathbb{Z}}$ of R -module homomorphisms; thus $\deg(f(x)) = \deg f + \deg x$. A *chain complex* over R is a pair (C, d) where C is a graded R -module and $d: C \rightarrow C$ is a map of degree -1 such that $d^2 = 0$. The map d is called the *differential* or *boundary operator* of C . We often suppress d from the notation and simply say that C is a chain complex. We define the *cycles* $Z(C)$, *boundaries* $B(C)$, and *homology* $H(C)$ by $Z(C) = \ker d$, $B(C) = \operatorname{im} d$, and $H(C) = Z(C)/B(C)$. These are all graded modules.

One often comes across graded modules C with an endomorphism d of square zero such that d has degree $+1$ instead of -1 . In this case it is customary to use superscripts instead of subscripts to denote the grading, so that $C = (C^n)_{n \in \mathbb{Z}}$ and $d = (d^n: C^n \rightarrow C^{n+1})$. Such a pair (C, d) is called a *cochain complex*. There is no essential difference between chain complexes and cochain complexes, since we can always convert one to the other by setting $C_n = C^{-n}$. We will therefore confine ourselves, for the most part, to discussing chain complexes, it being understood that everything applies to cochain complexes by reindexing as above. [Note, however, that there is a difference when we consider *non-negative* complexes, i.e., complexes such that C_n

[or C^n] = 0 for $n < 0$; if the differential is thought of as going from left to right, then a non-negative chain complex extends indefinitely to the left, whereas a non-negative cochain complex extends indefinitely to the right.] In discussing cochain complexes, one often prefixes “co” to much of the terminology; thus d may be called a coboundary operator, and we have cocycles $Z(C)$, coboundaries $B(C)$, and cohomology $H(C) = (H^n(C))_{n \in \mathbb{Z}}$.

If (C, d) and (C', d') are chain complexes, then a *chain map* from C to C' is a graded module homomorphism $f: C \rightarrow C'$ of degree 0 such that $d'f = fd$. A *homotopy* h from a chain map f to a chain map g is a graded module homomorphism $h: C \rightarrow C'$ of degree 1 such that $d'h + hd = f - g$. We write $f \simeq g$ and say that f is *homotopic* to g if there is a homotopy from f to g .

(0.1) Proposition. *A chain map $f: C \rightarrow C'$ induces a map $H(f): H(C) \rightarrow H(C')$, and $H(f) = H(g)$ if $f \simeq g$.* $\square$

The abelian group of homotopy classes of chain maps $C \rightarrow C'$ will be denoted $[C, C']$. It is often useful to interpret $[C, C']$ as the 0-th homology group of a certain “function complex” $\mathcal{H}om_R(C, C')$, defined as follows: $\mathcal{H}om_R(C, C')_n$ is the set of graded module homomorphisms of degree n from C to C' [thus $\mathcal{H}om_R(C, C')_n = \prod_{q \in \mathbb{Z}} \text{Hom}_R(C_q, C'_{q+n})$], and the boundary operator $D_n: \mathcal{H}om_R(C, C')_n \rightarrow \mathcal{H}om_R(C, C')_{n-1}$ is defined by $D_n(f) = d'f - (-1)^n fd$. [The sign here makes $D^2 = 0$. It is also consistent with other standard sign conventions, cf. exercise 3 below.] Note that the 0-cycles are precisely the chain maps $C \rightarrow C'$, and the 0-boundaries are the null-homotopic chain maps. Thus $H_0(\mathcal{H}om_R(C, C')) = [C, C']$. More generally, there is an interpretation of $H_n(\mathcal{H}om_R(C, C'))$ in terms of chain maps. Consider the complex $(\Sigma^n C, \Sigma^n d)$ defined by $(\Sigma^n C)_p = C_{p-n}$, $\Sigma^n d = (-1)^n d$; this complex is called the *n-fold suspension* of C . [If $n = 1$, we write ΣC instead of $\Sigma^1 C$.] Let $[C, C']_n = [\Sigma^n C, C']$. Then we have $H_n(\mathcal{H}om_R(C, C')) = [C, C']_n$. The elements of $[C, C']_n$ are called *homotopy classes of chain maps of degree n*.

A chain map $f: C \rightarrow C'$ is called a *homotopy equivalence* if there is a chain map $f': C' \rightarrow C$ such that $f'f \simeq \text{id}_C$ and $ff' \simeq \text{id}_{C'}$. And a chain map f is called a *weak equivalence* if $H(f): H(C) \rightarrow H(C')$ is an isomorphism.

(0.2) Proposition. *Any homotopy equivalence is a weak equivalence.* $\square$

A chain complex C is called *contractible* if it is homotopy equivalent to the zero complex, or, equivalently, if $\text{id}_C \simeq 0$. A homotopy from id_C to 0 is called a *contracting homotopy*. Any contractible chain complex is *acyclic*, i.e., $H(C) = 0$.

(0.3) Proposition. *C is contractible if and only if it is acyclic and each short exact sequence $0 \rightarrow Z_{n+1} \hookrightarrow C_{n+1} \xrightarrow{\bar{d}} Z_n \rightarrow 0$ splits, where $\bar{d}$ is induced by d .*

PROOF. If h is a contracting homotopy, then $(h|Z): Z \rightarrow C$ splits the surjection $\bar{d}: C \rightarrow Z$. Conversely, suppose we have a splitting $s: Z \rightarrow C$, whence a

graded module decomposition $C = \ker \bar{d} \oplus \operatorname{im} s = Z \oplus \operatorname{im} s$. We then get a contracting homotopy $h: C \rightarrow C$ by setting $h|Z = s$ and $h| \operatorname{im} s = 0$. $\square$

(0.4) Proposition. A short exact sequence $0 \rightarrow C' \xrightarrow{i} C \xrightarrow{\pi} C'' \rightarrow 0$ of chain complexes gives rise to a long exact sequence in homology:

$$\cdots \rightarrow H_n(C') \xrightarrow{H(i)} H_n(C) \xrightarrow{H(\pi)} H_n(C'') \xrightarrow{\partial} H_{n-1}(C') \rightarrow \cdots$$

The “connecting homomorphism” ∂ is natural, in the sense that a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & C' & \longrightarrow & C & \longrightarrow & C'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & E' & \longrightarrow & E & \longrightarrow & E'' \longrightarrow 0 \end{array}$$

with exact rows yields a commutative square

$$\begin{array}{ccc} H_n(C'') & \longrightarrow & H_{n-1}(C') \\ \downarrow & & \downarrow \\ H_n(E'') & \longrightarrow & H_{n-1}(E'). \end{array}$$

$\square$

(0.5) Corollary. The inclusion $i: C' \rightarrow C$ is a weak equivalence if and only if C'' is acyclic. $\square$

This shows that the cokernel C'' of i is the appropriate object to consider if we want to measure the “difference” between $H(C)$ and $H(C')$. We now wish to define a “homotopy-theoretic” cokernel for an arbitrary chain map $f: C' \rightarrow C$, which plays the same role as the cokernel in the case of an inclusion: The *mapping cone* of $f: (C', d') \rightarrow (C, d)$ is defined to be the complex (C'', d'') with $C'' = C \oplus \Sigma C'$ (as a graded module) and $d''(c, c') = (dc + fc', -d'c')$. In matrix notation, we have

$$d'' = \begin{pmatrix} d & f \\ 0 & \Sigma d' \end{pmatrix}.$$

See exercise 2 below for the motivation for this definition.

(0.6) Proposition. Let $f: C' \rightarrow C$ be a chain map with mapping cone C'' . There is a long exact homology sequence

$$\cdots \rightarrow H_n(C') \xrightarrow{H(f)} H_n(C) \rightarrow H_n(C'') \rightarrow H_{n-1}(C') \rightarrow \cdots$$

In particular, f is a weak equivalence if and only if C'' is acyclic.

PROOF. There is a short exact sequence $0 \rightarrow C \rightarrow C'' \rightarrow \Sigma C' \rightarrow 0$; now apply (0.4). By checking the definition of the connecting homomorphism $H_n(\Sigma C') \rightarrow H_{n-1}(C)$, one finds that it equals $H_{n-1}(f): H_{n-1}(C') \rightarrow H_{n-1}(C)$. $\square$

The mapping cone is also useful for studying homotopy equivalences, not just weak equivalences:

(0.7) Proposition. *A chain map $f: C' \rightarrow C$ is a homotopy equivalence if and only if its mapping cone C'' is contractible.*

PROOF. A straightforward computational proof can be found in the standard references (or can be supplied by the reader). For the sake of variety, we will sketch a conceptual proof. Suppose first that C'' is contractible. One then checks easily that the function complex $\mathcal{H}om_R(D, C'')$ is contractible for any complex D ; in particular, it is acyclic. One also checks that $\mathcal{H}om_R(D, C'')$ is isomorphic to the mapping cone of $\mathcal{H}om_R(D, f): \mathcal{H}om_R(D, C') \rightarrow \mathcal{H}om_R(D, C)$. It therefore follows from (0.6) that $\mathcal{H}om_R(D, f)$ is a weak equivalence. Looking at H_0 , we deduce that f induces an isomorphism $[D, C'] \rightarrow [D, C]$ for any D ; hence f is a homotopy equivalence by a standard argument. Conversely, suppose f is a homotopy equivalence. Then one shows easily that $\mathcal{H}om_R(D, f): \mathcal{H}om_R(D, C') \rightarrow \mathcal{H}om_R(D, C)$ is a homotopy equivalence, so its mapping cone $\mathcal{H}om_R(D, C'')$ is acyclic by 0.6. In particular, $[D, C''] = 0$ for any D , and this implies that C'' is contractible. $\square$

Finally, we recall briefly the Künneth and universal coefficient theorems. If (C, d) (resp. (C', d')) is a chain complex of right (resp. left) R -modules, then we define their *tensor product* $C \otimes_R C'$ by $(C \otimes_R C')_n = \bigoplus_{p+q=n} C_p \otimes_R C'_q$, with differential D given by $D(c \otimes c') = dc \otimes c' + (-1)^{\deg c} c \otimes d'c'$ for $c \in C$, $c' \in C'$. The sign here can be remembered by means of the following *sign convention*: When something of degree p is moved past something of degree q , the sign $(-1)^{pq}$ is introduced. [In the present case, the differential, which is of degree -1 , is moved past c , so we get the sign $(-1)^{-\deg c} = (-1)^{\deg c}$.] Note that $C \otimes_R C'$ is simply a complex of abelian groups for general R , but it is a complex of R -modules if R is commutative.

(0.8) Proposition (Künneth Formula). *Let R be a principal ideal domain and let C and C' be chain complexes such that C is dimension-wise free. There are natural exact sequences*

$$\begin{aligned} 0 \rightarrow \bigoplus_{p \in \mathbb{Z}} H_p(C) \otimes_R H_{n-p}(C') &\rightarrow H_n(C \otimes_R C') \\ &\rightarrow \bigoplus_{p \in \mathbb{Z}} \text{Tor}_1^R(H_p(C), H_{n-p-1}(C')) \rightarrow 0 \end{aligned}$$

and

$$\begin{aligned} 0 \rightarrow \prod_{p \in \mathbb{Z}} \text{Ext}_R^1(H_p(C), H_{p+n+1}(C')) &\rightarrow H_n(\mathcal{H}om_R(C, C')) \\ &\rightarrow \prod_{p \in \mathbb{Z}} \text{Hom}_R(H_p(C), H_{p+n}(C')) \rightarrow 0, \end{aligned}$$

and these sequences split. $\square$

We will not recall the definitions of Tor and Ext at this point, since we will be defining them in much greater generality in §III.2.

An important special case of 0.8 is that where C' consists of a single module M , regarded as a complex concentrated in dimension 0 (i.e. $C'_0 = M$, $C'_n = 0$ for $n \neq 0$). In this case 0.8 is called the *universal coefficient theorem*, and the exact sequences take the following form:

$$0 \rightarrow H_n(C) \otimes_R M \rightarrow H_n(C \otimes_R M) \rightarrow \text{Tor}_1^R(H_{n-1}(C), M) \rightarrow 0$$

and

$$0 \rightarrow \text{Ext}_R^1(H_{n-1}(C), M) \rightarrow H^n(\mathcal{H}om_R(C, M)) \rightarrow \text{Hom}_R(H_n(C), M) \rightarrow 0.$$

[Here we are following standard conventions in regarding $\mathcal{H}om_R(C, M)$ as a cochain complex, with $\mathcal{H}om_R(C, M)^n = \mathcal{H}om_R(C, M)_{-n} = \text{Hom}_R(C_n, M)$; the last equality comes from the fact that the only non-zero component of a graded map $f: C \rightarrow M$ of degree $-n$ is $f_n: C_n \rightarrow M$.]

EXERCISES

1. Let $T: (R\text{-modules}) \rightarrow (S\text{-modules})$ be a covariant functor which takes the zero module to the zero module (or, equivalently, which takes zero maps to zero maps). For any chain complex (C, d) over R , there is then a chain complex (TC, Td) over S . If T is an exact functor, show that $H(TC) \approx TH(C)$. [Recall that T is *exact* if it carries exact sequences to exact sequences. It follows that T preserves injections, surjections, kernels, cokernels, etc.]
2. (Motivation for the definition of the mapping cone) Given a chain map $f: C' \rightarrow C$, the “homotopy theoretic cokernel” of f should fit into a diagram $C' \xrightarrow{f} C \xrightarrow{g} C''$ with $gf \simeq 0$. Thus C'' must receive a chain map g [of degree 0] from C and a homotopy h [of degree 1] from C' ; this suggests setting $C'' = C \oplus \Sigma C'$ as a graded module, and taking g and h to be inclusions. Show that the definition of the boundary operator d'' is now forced on us by the requirement that g be a chain map and that h be a homotopy from gf to 0. [Using matrix notation, set

$$d'' = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

and note that

$$g = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad h = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Now write out the matrix equations $d''g = gd$ and $d''h + hd' = gf$ and solve for $\alpha, \beta, \gamma, \delta$.]

The remaining exercises are designed to illustrate various compatibility properties of our sign conventions in $\mathcal{H}om(-, -)$ and $- \otimes -$. Few readers will have the patience to do all of them, but you should at least do enough to convince yourself that all reasonable identities involving $\mathcal{H}om$ and $\otimes$ are true, provided one follows the sign convention stated above.

3. Given $u \in \mathcal{H}om_R(C, C')_p$ and $x \in C_q$, set $\langle u, x \rangle = u(x) \in C'_{p+q}$. Let d, d' , and D be the differentials in C, C' , and $\mathcal{H}om_R(C, C')$, respectively. Verify that $d'\langle u, x \rangle = \langle Du, x \rangle + (-1)^p \langle u, dx \rangle$. [In fact, this is nothing but a restatement of the definition of D , but it is a convenient form in which to remember that definition.] In other words, the evaluation map $\mathcal{H}om_R(C, C') \otimes C \rightarrow C'$, given by $u \otimes x \mapsto \langle u, x \rangle$, is a chain map.
4. Given $v \in \mathcal{H}om_R(C, C')_q$ and $u \in \mathcal{H}om_R(C', C'')_p$, their composite $u \circ v$ is in

$$\mathcal{H}om_R(C, C'')_{p+q}.$$

Verify that $D(u \circ v) = Du \circ v + (-1)^p u \circ Dv$, where D denotes the differential in any of the three $\mathcal{H}om$ complexes; in other words, composition of graded maps defines a chain map $\mathcal{H}om_R(C', C'') \otimes \mathcal{H}om_R(C, C') \rightarrow \mathcal{H}om_R(C, C'')$. [Hint: This and the remaining exercises are most conveniently done by taking the definition of D in the form given in exercise 3. From this point of view, one starts with the equation $\langle u \circ v, x \rangle = \langle u, \langle v, x \rangle \rangle$ ($x \in C$), which is the definition of composition, and one applies d'' to both sides. Applying exercise 3 several times, one obtains

$$\begin{aligned} \langle D(u \circ v), x \rangle + (-1)^{p+q} \langle u \circ v, dx \rangle &= \langle Du, \langle v, x \rangle \rangle + (-1)^p \langle u, \langle Dv, x \rangle \rangle \\ &\quad + (-1)^{p+q} \langle u, \langle v, dx \rangle \rangle \end{aligned}$$

which simplifies to

$$\langle D(u \circ v), x \rangle = \langle Du \circ v, x \rangle + (-1)^p \langle u \circ Dv, x \rangle,$$

as required.]

5. If C and C' are chain complexes over a commutative ring R , there is an isomorphism of graded modules $C \otimes_R C' \xrightarrow{\sim} C' \otimes_R C$ given by $c \otimes c' \mapsto (-1)^{\deg c \cdot \deg c'} c' \otimes c$. Prove that this isomorphism is a chain map.
6. Let C be a complex of right R -modules, C' a complex of left R -modules, and C'' a complex of $\mathbb{Z}$ -modules.

(a) There is a canonical isomorphism of graded abelian groups

$$\varphi: \mathcal{H}om_{\mathbb{Z}}(C \otimes_R C', C'') \xrightarrow{\sim} \mathcal{H}om_R(C, \mathcal{H}om_{\mathbb{Z}}(C', C'')),$$

given by $\langle \langle \varphi(u), c \rangle, c' \rangle = \langle u, c \otimes c' \rangle$ for $u \in \mathcal{H}om_{\mathbb{Z}}(C \otimes_R C', C'')$, $c \in C$, $c' \in C'$. [Sketch of proof: An element of $\mathcal{H}om_{\mathbb{Z}}(C \otimes_R C', C'')$ is a family of $\mathbb{Z}$ -module maps $C_p \otimes_R C'_q \rightarrow C''_{p+q+n}$. In view of the universal mapping property of the tensor product, this is the same as a family of R -module maps $C_p \rightarrow \text{Hom}_{\mathbb{Z}}(C'_q, C''_{p+q+n})$, i.e., as a graded R -module map $C \rightarrow \mathcal{H}om_{\mathbb{Z}}(C', C'')$ of degree n . Note here that $\mathcal{H}om_{\mathbb{Z}}(C', C'')$ is indeed a complex of right R -modules via the left action of R on C' and the contravariance of Hom in the first variable: $(ur)(c') = u(rc')$ for $u \in \mathcal{H}om_{\mathbb{Z}}(C', C'')$, $r \in R$, $c' \in C'$.] Show that φ is a chain map. [Hint: Let d, d' , and d'' be the differentials in C, C' , and C'' , and let D be the differential in any of the $\mathcal{H}om$ complexes under consideration. Apply d'' to both sides of the equation defining φ ; using exercise 3 several times, you should obtain

$$\langle \langle D(\varphi(u)), c \rangle, c' \rangle + (-1)^p \langle \langle \varphi(u), dc \rangle, c' \rangle + (-1)^{p+q} \langle \langle \varphi(u), c \rangle, d'c' \rangle$$

on the left, where $p = \deg u$ and $q = \deg c$, and

$$\langle Du, c \otimes c' \rangle + (-1)^p \langle u, dc \otimes c' \rangle + (-1)^{p+q} \langle u, c \otimes d'c' \rangle$$

on the right. Remembering the definition of φ , you can conclude that

$$\langle \langle D(\varphi(u)), c \rangle, c' \rangle = \langle Du, c \otimes c' \rangle$$

and hence that $\langle \langle D(\varphi(u)), c \rangle, c' \rangle = \langle \langle \varphi(Du), c \rangle, c' \rangle$]

(b) Deduce from (a) [or check directly] that a map $u: C \otimes_R C' \rightarrow C''$ of degree 0 is a chain map iff the corresponding map $C \rightarrow \mathcal{H}om_{\mathbb{Z}}(C', C'')$ is a chain map.

7. Let C and C' (resp. E and E') be complexes of right (resp. left) R -modules. Given $u \in \mathcal{H}om_R(C, C')$ and $v \in \mathcal{H}om_R(E, E')$, their tensor product

$$u \otimes v \in \mathcal{H}om_{\mathbb{Z}}(C \otimes_R E, C' \otimes_R E')$$

is defined by

$$\langle u \otimes v, c \otimes e \rangle = (-1)^{\deg v \cdot \deg c} \langle u, c \rangle \otimes \langle v, e \rangle \quad \text{for } c \in C, e \in E.$$

(a) Let D be the differential in any of the three $\mathcal{H}om$ complexes under consideration. Verify that $D(u \otimes v) = Du \otimes v + (-1)^{\deg u} u \otimes Dv$. In other words, the operation “tensor product of graded maps” defines a chain map

$$\mathcal{H}om_R(C, C') \otimes \mathcal{H}om_R(E, E') \rightarrow \mathcal{H}om_{\mathbb{Z}}(C \otimes_R E, C' \otimes_R E').$$

[Hint: Once again, you are advised to do this by differentiating both sides of the equation defining $u \otimes v$.]

(b) Deduce from (a) (or check directly) that if $u: C \rightarrow C'$ is a chain map (of degree 0), then there is a chain map $\mathcal{H}om_R(E, E') \rightarrow \mathcal{H}om_{\mathbb{Z}}(C \otimes_R E, C' \otimes_R E')$ given by $v \mapsto u \otimes v$.

(c) Deduce from (a) (or check directly) that the tensor product of chain maps is compatible with homotopy, in the following sense: Given chain maps and homotopies $u_0 \simeq u_1: C \rightarrow C'$ and $v_0 \simeq v_1: E \rightarrow E'$, one has $u_0 \otimes v_0 \simeq u_1 \otimes v_1: C \otimes_R E \rightarrow C' \otimes_R E'$. [Hint: u_0 and u_1 are homologous 0-cycles in $\mathcal{H}om_R(C, C')$, and similarly for v_0 and v_1 ; it now follows from the boundary formula in (a) that $u_0 \otimes v_0$ and $u_1 \otimes v_1$ are homologous 0-cycles in $\mathcal{H}om_{\mathbb{Z}}(C \otimes_R E, C' \otimes_R E')$.]

8. Prove that the tensor product operation of the previous exercise is compatible with composition of graded maps, i.e., that

$$(u \otimes v) \circ (u' \otimes v') = (-1)^{\deg v \cdot \deg u'} (u \circ u') \otimes (v \circ v')$$

whenever the composites are defined.

1 Free Resolutions

Let R be a ring (associative, with identity) and M a (left) R -module. A *resolution* of M is an exact sequence of R -modules

$$\cdots \rightarrow F_2 \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \xrightarrow{\varepsilon} M \rightarrow 0.$$

If each F_i is free, then this is called a *free resolution*. Free resolutions exist for any module M and can be constructed by an obvious step-by-step

procedure: Choose a surjection $\varepsilon: F_0 \rightarrow M$ with F_0 free, then choose a surjection $F_1 \rightarrow \ker \varepsilon$ with F_1 free, etc. Note that the initial segment $F_1 \rightarrow F_0 \rightarrow M \rightarrow 0$ of a free resolution is simply a *presentation* of M by generators and relations.

There are two useful ways to interpret a resolution in terms of chain complexes.¹ The first is to regard the resolution itself as an acyclic chain complex, with M in degree -1 . We will refer to this as the *augmented chain complex* associated to the resolution. The second way is to consider the non-negative chain complex $F = (F_i, \partial_i)_{i \geq 0}$ and to view $\varepsilon: F \rightarrow M$ as a chain map, where M is regarded as a chain complex concentrated in dimension 0. The exactness hypothesis, from this point of view, simply says that ε is a weak equivalence.

In case there is an integer n such that $F_i = 0$ for $i > n$, we will say that the resolution has *length* $\leq n$. In this case we will simply write

$$0 \rightarrow F_n \rightarrow \cdots \rightarrow F_0 \rightarrow M \rightarrow 0,$$

it being understood that the resolution continues to the left with zeroes.

EXAMPLES

1. A free module F admits the free resolution

$$0 \rightarrow F \xrightarrow{\text{id}} F \rightarrow 0$$

of length 0.

2. If $R = \mathbb{Z}$ (or any principal ideal domain) then submodules of a free module are free, hence any module M admits a free resolution

$$0 \rightarrow F_1 \rightarrow F_0 \rightarrow M \rightarrow 0$$

of length ≤ 1 . For example, the $\mathbb{Z}$ -module $\mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z}$ admits the resolution

$$0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}_2 \rightarrow 0.$$

3. Let M again be $\mathbb{Z}_2$, regarded now as a module over the polynomial ring $R = \mathbb{Z}[T]$, with T acting as 0 (i.e., $f(T)$ acts as multiplication by $f(0)$). Then M can be regarded as the quotient of R by the ideal generated by T and 2. Using unique factorization in R and the fact that T and 2 are relatively prime, one easily obtains the free resolution of length 2

$$0 \rightarrow R \xrightarrow{\partial_2} R \oplus R \xrightarrow{\partial_1} R \xrightarrow{\varepsilon} \mathbb{Z}_2 \rightarrow 0$$

where $\varepsilon(f) = f(0) \bmod 2$, and ∂_1 and ∂_2 are given by the matrices $(T \ 2)$ and $(-\frac{2}{T})$, respectively.

4. Let $R = \mathbb{Z}[T]/(T^2 - 1)$ and let t be the image of T in R . Let M be the R -module $R/(t - 1)$. (Equivalently, $M = \mathbb{Z}$, with t acting as the identity.)

¹ See §0 for terminology and basic facts concerning chain complexes.

Since $T^2 - 1 = (T - 1)(T + 1)$, it is clear that an element of R is annihilated by $t - 1$ (resp. $t + 1$) if and only if it is divisible by $t + 1$ (resp. $t - 1$). One therefore has a free resolution

$$\cdots R \xrightarrow{t-1} R \xrightarrow{t+1} R \xrightarrow{t-1} R \rightarrow M \rightarrow 0.$$

Note that, unlike the previous examples, this resolution is of *infinite* length. We will see, in fact, that M admits no free resolution of finite length, cf. §II.3, exercise 2.

In this book we will be primarily interested in the case where R is a group ring, so we digress now to recall what a group ring is.

2 Group Rings

Let G be a group, written multiplicatively. Let $\mathbb{Z}G$ (or $\mathbb{Z}[G]$) be the free $\mathbb{Z}$ -module generated by the elements of G . Thus an element of $\mathbb{Z}G$ is uniquely expressible in the form $\sum_{g \in G} a(g)g$, where $a(g) \in \mathbb{Z}$ and $a(g) = 0$ for almost all g . The multiplication in G extends uniquely to a $\mathbb{Z}$ -bilinear product $\mathbb{Z}G \times \mathbb{Z}G \rightarrow \mathbb{Z}G$; this makes $\mathbb{Z}G$ a ring, called the *integral group ring* of G .

Note that G is a subgroup of the group $(\mathbb{Z}G)^*$ of units of $\mathbb{Z}G$ and that we have the following *universal mapping property*:

(2.1) Given a ring R and a group homomorphism $f: G \rightarrow R^*$, there is a unique extension of f to a ring homomorphism $\mathbb{Z}G \rightarrow R$. Thus we have the “adjunction formula”

$$\text{Hom}_{(\text{rings})}(\mathbb{Z}G, R) \approx \text{Hom}_{(\text{groups})}(G, R^*).$$

EXAMPLES

1. Let G be cyclic of order n and let t be a generator. Then the powers t^i ($0 \leq i \leq n - 1$) form a $\mathbb{Z}$ -basis for $\mathbb{Z}G$, and one has $t^n = 1$. It follows that $\mathbb{Z}G \approx \mathbb{Z}[T]/(T^n - 1)$.

2. If G is infinite cyclic with generator t , then $\mathbb{Z}G$ has a $\mathbb{Z}$ -basis $\{t^i\}_{i \in \mathbb{Z}}$. Hence $\mathbb{Z}G$ can be identified with the ring (usually denoted $\mathbb{Z}[t, t^{-1}]$) of Laurent polynomials $\sum_{i \in \mathbb{Z}} a_i t^i$ ($a_i \in \mathbb{Z}$, $a_i = 0$ for almost all i).

EXERCISES

1. For any group G we define the *augmentation map* to be the ring homomorphism $\varepsilon: \mathbb{Z}G \rightarrow \mathbb{Z}$ such that $\varepsilon(g) = 1$ for all $g \in G$. The kernel of ε is called the *augmentation ideal* of $\mathbb{Z}G$ and is denoted I or IG .

(a) Show that the elements $g - 1$ ($g \in G$, $g \neq 1$) form a basis for I as a $\mathbb{Z}$ -module.

(b) If S is a set of generators for G , show that the elements $s - 1$ ($s \in S$) generate I as a left ideal.

- (c) Conversely, if S is a set of elements of G such that the elements $s - 1$ ($s \in S$) generate I as a left ideal, show that S generates G . [Hint: The hypothesis on S implies that every element of I is a sum of elements of the form $g - g'$, where $g = g's^{\pm 1}$ for some $s \in S$. Writing $g - 1$ in this way, where $g \in G$ is arbitrary, deduce that there is a sequence $g_1, g_2, \dots, g_n$ such that $g_1 = g$, $g_n = 1$, and $g_i = g_{i+1}s_i^{\pm 1}$ for some $s_i \in S$. See exercise 2 of §3 below for an alternative proof.]
- (d) Show that G is a finitely generated group if and only if I is finitely generated as a left ideal.
2. Let G be cyclic with generator t and let M be the G -module $\mathbb{Z}$, with t acting as the identity. (Equivalently, $M = \mathbb{Z}G/I = \mathbb{Z}G/(t - 1)$.) Write down a free resolution of M . [See §1, example 4, for the case where $|G| = 2$.]

3 G -modules

A (left) $\mathbb{Z}G$ -module, also called a G -module, consists of an abelian group A together with a homomorphism from $\mathbb{Z}G$ to the ring of endomorphisms of A . In view of 2.1, such a ring homomorphism corresponds to a group homomorphism from G to the group of automorphisms of A . Thus a G -module is simply an abelian group A together with an action of G on A . For example, one has for any A the *trivial* module structure, with $ga = a$ for $g \in G$, $a \in A$. (Thus $ra = \varepsilon(r) \cdot a$ for $r \in \mathbb{Z}G$, where $\varepsilon: \mathbb{Z}G \rightarrow \mathbb{Z}$ is the augmentation map discussed in §2, exercise 1.)

One way of constructing G -modules is by linearizing permutation representations. More precisely, if X is a G -set (i.e., a set with G -action), then one forms the free abelian group $\mathbb{Z}X$ (also denoted $\mathbb{Z}[X]$) generated by X and one extends the action of G on X to a $\mathbb{Z}$ -linear action of G on $\mathbb{Z}X$. The resulting G -module is called a *permutation module*. In particular, one has a permutation module $\mathbb{Z}[G/H]$ for every subgroup H of G , where G/H is the set of cosets gH and G acts on G/H by left translation.

The operation of disjoint union in the category of G -sets corresponds to the direct sum operation in the category of G -modules:

$$\mathbb{Z}[\coprod X_i] = \oplus \mathbb{Z}X_i.$$

It follows that every permutation module $\mathbb{Z}X$ admits a decomposition

$$\mathbb{Z}X \approx \oplus \mathbb{Z}[G/G_x],$$

where x ranges over a set of representatives for the G -orbits in X and G_x is the isotropy subgroup of G at x . In particular, if X is a *free* G -set (i.e., if all isotropy groups are trivial), then $G/G_x = G$, and we obtain:

(3.1) Proposition. *Let X be a free G -set and let E be a set of representatives for the G -orbits in X . Then $\mathbb{Z}X$ is a free $\mathbb{Z}G$ -module with basis E .*

Remark. If k is an arbitrary commutative ring, then one can form the *group algebra* kG of G over k . This is the free k -module generated by G , with the unique k -bilinear product extending the group multiplication on G . Everything we have done in this section generalizes in an obvious way to this situation. For example, a kG -module is simply a k -module A together with a (k -linear) action of G on A .

EXERCISES

1. Let H be a subgroup of G and let E be a set of representatives for the right cosets Hg . Show that $\mathbb{Z}G$, regarded as a left-module over its subring $\mathbb{Z}H$, is free with basis E .
2. Use permutation modules to give a non-computational solution of exercise 1c of §2. [Let H be the subgroup of G generated by S and consider $\mathbb{Z}[G/H]$. It has an element fixed by H and hence annihilated by I . But then the element is fixed by G .]

4 Resolutions of $\mathbb{Z}$ over $\mathbb{Z}G$ via Topology

In this section we will consider $\mathbb{Z}$ as a G -module with trivial G -action, and we will see that free resolutions of this module arise from free actions of G on contractible complexes.

By a G -complex we will mean a CW -complex X together with an action of G on X which permutes the cells. Thus we have for each $g \in G$ a homeomorphism $x \mapsto gx$ of X such that the image $g\sigma$ of any cell σ of X is again a cell. For example, if X is a simplicial complex on which G acts simplicially, then X is a G -complex.

If X is a G -complex then the action of G on X induces an action of G on the cellular chain complex $C_*(X)$, which thereby becomes a chain complex of G -modules. Moreover, the canonical augmentation $\varepsilon: C_0(X) \rightarrow \mathbb{Z}$ (defined by $\varepsilon(v) = 1$ for every 0-cell v of X) is a map of G -modules.

We will say that X is a *free G -complex* if the action of G freely permutes the cells of X (i.e., $g\sigma \neq \sigma$ for all σ if $g \neq 1$). In this case each chain module $C_n(X)$ has a $\mathbb{Z}$ -basis which is freely permuted by G , hence by 3.1 $C_n(X)$ is a *free $\mathbb{Z}G$ -module* with one basis element for every G -orbit of cells. (Note that to obtain a specific basis we must choose a representative cell from each orbit and we must choose an orientation of each such representative.)

Finally, if X is contractible, then $H_*(X) \approx H_*(\text{pt.})$; in other words, the sequence

$$\cdots \rightarrow C_n(X) \xrightarrow{\partial} C_{n-1}(X) \rightarrow \cdots \rightarrow C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0$$

is exact. We have, therefore:

(4.1) Proposition. *Let X be a contractible free G -complex. Then the augmented cellular chain complex of X is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$.*

The reader who has studied covering spaces has, of course, seen many examples of free G -complexes. Indeed, suppose $p: \tilde{Y} \rightarrow Y$ is a regular covering map with G as group of deck transformations. (See the appendix to this chapter for a review of regular covers.) If Y is a CW -complex, then it is an elementary fact that $\tilde{Y}$ inherits a CW -structure such that the G -action permutes the cells, cf. Schubert [1968], III.6.9. Explicitly, the open cells of $\tilde{Y}$ lying over an open cell σ of Y are simply the connected components of $p^{-1}\sigma$; these cells are permuted freely and transitively by G , and each is mapped homeomorphically onto σ under p . Thus $\tilde{Y}$ is a free G -complex and $C_*\tilde{Y}$ is a complex of free $\mathbb{Z}G$ -modules with one basis element for each cell of Y .

In view of 4.1, it is natural now to consider CW -complexes Y satisfying the following three conditions:

- (i) Y is connected.
- (ii) $\pi_1 Y = G$.
- (iii) The universal cover X of Y is contractible.

Such a Y is called an *Eilenberg–MacLane complex* of type $(G, 1)$, or simply a $K(G, 1)$ -complex. Condition (ii) above is to be interpreted as meaning that we are given a basepoint $y \in Y$ and a specific isomorphism $\pi_1(Y, y) \xrightarrow{\sim} G$ by which we identify $\pi_1(Y, y)$ with G .

It is sometimes convenient to note that condition (iii) above can be replaced by:

- (iii') $H_i X = 0$ for $i \geq 2$.

Indeed, we clearly have (iii) $\Rightarrow$ (iii'). Conversely, (iii') implies that X is *acyclic*, i.e., that $H_* X \approx H_*(\text{pt.})$; and it is shown in elementary homotopy theory that simply-connected acyclic CW -complexes are contractible. [The reader who is not familiar with this fact can simply take (i), (ii), and (iii') as the *definition* of a $K(G, 1)$ -complex, since we will never make serious use of the fact that X is contractible. Note, for instance, that 4.1 remains true if “contractible” is replaced by “acyclic.”]

It also follows from elementary homotopy theory that (iii) can be replaced by

- (iii'') $\pi_i Y = 0$ for $i \geq 2$.

Thus the Eilenberg–MacLane complexes are precisely the *aspherical* complexes studied by Hurewicz (cf. Introduction). Once again, we will not need to use this fact.

If Y is a $K(G, 1)$ then the universal cover $p: X \rightarrow Y$ is a regular cover whose group is isomorphic to $\pi_1 Y = G$. [This depends on a choice of basepoint in X , cf. Appendix.] We therefore obtain from 4.1:

(4.2) Proposition. *If Y is a $K(G, 1)$ then the augmented cellular chain complex of the universal cover of Y is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$.*

We now give one example. There will be many other examples in Chapter II and later in the book.

(4.3) EXAMPLE. Let $G = F(S)$, the free group generated by a set S . Let $Y = \bigvee_{s \in S} S_s^1$, a bouquet of circles S_s^1 in 1-1 correspondence with S . Thus Y is a 1-dimensional CW-complex with exactly one vertex and with one 1-cell for each element of S . One knows that $\pi_1 Y \approx F(S)$. Moreover, Y is a $K(F(S), 1)$ since condition (iii') above holds for trivial reasons. [Y is 1-dimensional.] As basepoint in Y we take a vertex x_0 ; this then represents the unique G -orbit of vertices of Y and hence generates the free $\mathbb{Z}G$ -module $C_0(Y)$. As basis for $C_1(Y)$ we take, for each $s \in S$, an oriented 1-cell e_s of Y which lies over S_s^1 (traversed in the positive sense). Replacing e_s by ge_s for suitable $g \in G$, we can assume that the initial vertex of e_s is the basepoint x_0 ; the endpoint of e_s is then sx_0 (cf. Appendix, A1) so that $\partial e_s = sx_0 - x_0 = (s - 1)x_0$. We obtain, therefore, the free resolution

$$(4.4) \quad 0 \rightarrow \mathbb{Z}G^{(S)} \xrightarrow{\partial} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0$$

where $\mathbb{Z}G^{(S)}$ is a free $\mathbb{Z}G$ -module with basis $(e_s)_{s \in S}$, $\partial e_s = s - 1$, and $\varepsilon(g) = 1$ for all $g \in G$.

Remarks

1. Note that ε coincides with the natural augmentation of the group ring $\mathbb{Z}G$, as defined in §2, exercise 1. The exactness of 4.4 says, therefore, that the augmentation ideal of $\mathbb{Z}[F(S)]$ is a free left $\mathbb{Z}[F(S)]$ -module with basis $(s - 1)_{s \in S}$. [We will later indicate a purely algebraic proof of this fact, cf. §IV.2, exercise 3b.] The reader who is not used to working with modules over non-commutative rings may find it surprising that $\mathbb{Z}G$, the free module of rank 1, can contain a submodule which is free of rank > 1 ; this sort of thing cannot happen over a commutative ring R , since then any two elements $a, b \in R$ are linearly dependent: $(-b) \cdot a + a \cdot b = 0$.

2. If S contains a single element t (i.e., G is infinite cyclic) then (4.4) reduces to the "obvious" resolution which the reader probably wrote down in §2, exercise 2:

$$(4.5) \quad 0 \rightarrow \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0.$$

Note that X , in this case, is simply $\mathbb{R}^1$, with t acting by $x \mapsto x + 1$:

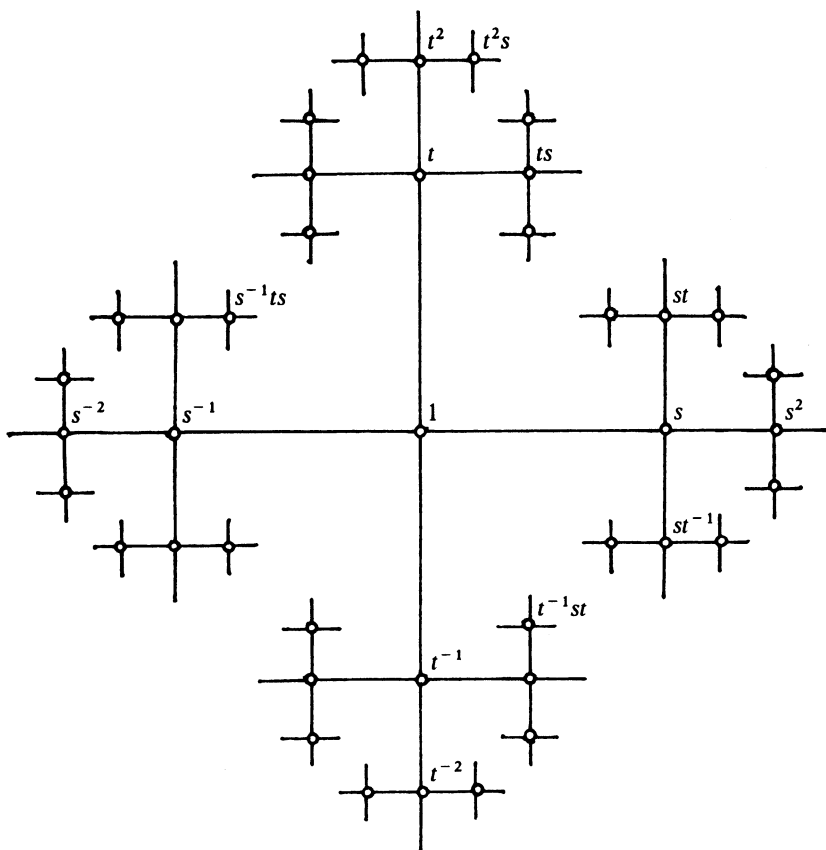
$$\begin{array}{ccccccc} & \circ & & \circ & & \circ & & \circ & & \circ \\ & | & & | & & | & & | & & | \\ t^{-2}x_0 & & t^{-1}x_0 & & x_0 & & tx_0 & & t^2x_0 \end{array}$$

3. The contractible free $F(S)$ -complex X , and hence the resolution 4.4, can easily be constructed directly, without appeal to the theory of covering complexes, or to the fact that $\pi_1(\bigvee S^1) \approx F(S)$. Namely, X can be defined as the 1-dimensional simplicial complex whose vertices are the elements of $G = F(S)$ and whose 1-simplices are the pairs $\{g, gs\}$ ($g \in G, s \in S$). The action

of G on itself by left translation induces a simplicial action of G on X , which is easily seen to be free. Finally, one can construct an explicit contracting homotopy which contracts X to the vertex 1 along paths of the form

$$\begin{array}{ccccccc} \circ & \text{---} & \circ & \cdots & \circ & \text{---} & \circ \\ g = g_n & & g_{n-1} & & g_1 & & g_0 = 1 \end{array}$$

where $g = s_1^{\varepsilon_1} \cdots s_n^{\varepsilon_n}$ is a reduced word of length n ($s_i \in S$, $\varepsilon_i = \pm 1$, $\varepsilon_i = \varepsilon_{i+1}$ if $s_i = s_{i+1}$) and $g_i = s_1^{\varepsilon_1} \cdots s_i^{\varepsilon_i}$ ($0 \leq i \leq n$). In case S is a two-element set $\{s, t\}$, for example, X is the tree pictured below:



EXERCISES

1. If X is an arbitrary G -complex, is $C_n(X)$ necessarily a permutation module?
- *2. Let X be a free G -complex, where G is an arbitrary group. Show that every point of X has a neighborhood U such that $gU \cap U = \emptyset$ for all $g \neq 1$ in G . Deduce that $X \rightarrow X/G$ is a regular covering map with G as group of covering transformations. In particular, if X is contractible then X/G is a $K(G, 1)$ and X is its universal cover.

[Hint: A CW -complex X , with explicitly given characteristic maps $(B^n, S^{n-1}) \rightarrow (\bar{\sigma}, \partial\sigma)$ for its cells, admits a canonical open cover $\{U_\sigma\}$ indexed by the cells, such that $U_\sigma \cap U_\tau = \emptyset$ if σ and τ are distinct cells of the same dimension. If X is simplicial, for example, we can take U_σ to be the open star of the barycenter of σ in the barycentric subdivision of X .]

3. Let G be a free abelian group of rank 2. Use the methods of this section to construct a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$.

*4. (This exercise requires some homotopy theory.)

(a) For any group G , show that there exists a $K(G, 1)$. [Start with a connected 2-complex with $\pi_1 = G$; then attach cells to kill the higher homotopy.]

(b) Show that the construction in (a) can be made functorial in G . [Given the $(n-1)$ -skeleton Y^{n-1} , attach an n -cell for every possible map $S^{n-1} \rightarrow Y^{n-1}$.]

5 The Standard Resolution

Let X be a contractible simplicial complex on which a group G acts simplicially. It may happen that the G -action is free on the vertices but not on the higher-dimensional simplices. [Note: This cannot happen if G is torsion-free.] In this case we can still obtain a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ by using, instead of the usual chain complex $C_*(X)$, the *ordered* chain complex $C'_*(X)$ (cf. Spanier [1966], ch. 4, §3). Thus $C'_n(X)$ has a $\mathbb{Z}$ -basis consisting of the ordered $(n+1)$ -tuples $(v_0, \dots, v_n)$ of vertices of X such that $\{v_0, \dots, v_n\}$ is a simplex of X (of dimension $< n$ if the v_i are not all distinct). Since G clearly acts freely on these $(n+1)$ -tuples, we obtain a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$.

The most obvious example of such an X is the “simplex” spanned by G ; i.e., the vertices of X are the elements of G (with G acting by left translation), and every finite subset of G is a simplex of X . [If G is finite, this really is a simplex; otherwise it is an infinite dimensional analogue. In any case it is contractible by a straight-line contracting homotopy.] The corresponding free resolution $F_* = C'_*(X)$ is called the *standard resolution* of $\mathbb{Z}$ over $\mathbb{Z}G$. Explicitly, F_n is the free $\mathbb{Z}$ -module generated by the $(n+1)$ -tuples $(g_0, \dots, g_n)$ of elements of G , with the G -action given by $g \cdot (g_0, \dots, g_n) = (gg_0, \dots, gg_n)$. The boundary operator $\partial: F_n \rightarrow F_{n-1}$ is given by $\partial = \sum_{i=0}^n (-1)^i d_i$, where

$$(5.1) \quad d_i(g_0, \dots, g_n) = (g_0, \dots, \hat{g}_i, \dots, g_n).$$

The augmentation $\varepsilon: F_0 \rightarrow \mathbb{Z}$ is given by $\varepsilon(g_0) = 1$. Note that the acyclicity (i.e., exactness) of this augmented complex has been deduced from the contractibility of X , but one can also verify it directly by writing down a contracting homotopy $h: F_n \rightarrow F_{n+1}$ for the underlying augmented complex of $\mathbb{Z}$ -modules (i.e., h will not be a map of G -modules). Namely, we define h by $h(g_0, \dots, g_n) = (1, g_0, \dots, g_n)$ if $n \geq 0$ and $h(1) = (1)$ if $n = -1$.

As basis for the free $\mathbb{Z}G$ -module F_n we may take the $(n + 1)$ -tuples whose first element is 1, since these represent the G -orbits of $(n + 1)$ -tuples. It is often convenient to write such an $(n + 1)$ -tuple in the form $(1, g_1, g_1g_2, \dots, g_1g_2 \cdots g_n)$ and to introduce the “bar notation”

$$[g_1|g_2|\cdots|g_n] = (1, g_1, g_1g_2, \dots, g_1g_2 \cdots g_n).$$

(If $n = 0$ there is only one such basis element, denoted $[\]$; if we identify F_0 with $\mathbb{Z}G$ in the obvious way, then $[\] = 1$.) It is easy to compute $\partial: F_n \rightarrow F_{n-1}$ in terms of this $\mathbb{Z}G$ -basis $\{[g_1|\cdots|g_n]\}$; one finds $\partial = \sum_{i=0}^n (-1)^i d_i$, where d_i is the $\mathbb{Z}G$ -homomorphism given by

$$(5.2) \quad d_i[g_1|\cdots|g_n] = \begin{cases} g_1[g_2|\cdots|g_n] & i = 0 \\ [g_1|\cdots|g_{i-1}|g_i g_{i+1}|g_{i+2}|\cdots|g_n] & 0 < i < n \\ [g_1|\cdots|g_{n-1}] & i = n. \end{cases}$$

This standard resolution F_* is often called the *bar resolution*. In low dimensions it has the form

$$F_2 \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0,$$

where $\partial_2([g|h]) = g[h] - [gh] + [g]$, $\partial_1([g]) = g[\] - [\] = g - 1$, and $\varepsilon(1) = 1$.

Finally, we mention the *normalized standard (or bar) resolution* $\bar{F}_* = F_*/D_*$, where D_* , the “degenerate” subcomplex of F_* , is generated by the elements $(g_0, \dots, g_n)$ such that $g_i = g_{i+1}$ for some i . In terms of the bar notation, D_* can be described as the G -subcomplex of F_* generated by the elements $[g_1|\cdots|g_n]$ such that $g_i = 1$ for some i . Thus $\bar{F}_n$ is a free $\mathbb{Z}G$ -module with one basis element (still denoted $[g_1|\cdots|g_n]$) for every n -tuple of *non-trivial* elements of G . The fact that $\bar{F}_*$ is acyclic over $\mathbb{Z}$, and hence a resolution, follows from general facts about normalization (cf. MacLane [1963], VIII.6); alternatively, one can simply observe that the contracting homotopy h defined above carries D_* into itself and hence induces a contracting homotopy of the quotient $\bar{F}_*$.

EXERCISES

1. Write down the homotopy operator h in terms of the $\mathbb{Z}$ -basis $g[g_1|\cdots|g_n]$ for F_n .
2. Write out the normalized bar resolution in case G is the group of order 2; compare it with the resolution given in §1, example 4.
- *3. (a) Show that the standard resolution is the cellular chain complex of a free contractible G -complex X . Note that this reproves the result of exercises 4a and 4b of §4. [For each $(n + 1)$ -tuple $\sigma = (g_0, \dots, g_n)$, let Δ_σ be a copy of the standard n -simplex with vertices $v_0, \dots, v_n$. Let $d_i\sigma = (g_0, \dots, \hat{g}_i, \dots, g_n)$ and let $\delta_i: \Delta_{d_i\sigma} \rightarrow \Delta_\sigma$ be the linear embedding which sends $v_0, \dots, v_{n-1}$ to $v_0, \dots, \hat{v}_i, \dots, v_n$. Form the disjoint union $X_0 = \coprod_\sigma \Delta_\sigma$ and obtain X by identifying $\Delta_{d_i\sigma}$ with its image under δ_i

for all σ and all i . Use the quotient map $C(X_0) \rightarrow C(X)$ to compute the boundary operator on $C(X)$ and see that $C(X) \approx F$. To prove X contractible, consider $h\sigma = (1, g_0, \dots, g_n)$ for each σ , and use the straight-line homotopy between $\delta_0: \Delta_\sigma \rightarrow \Delta_{h\sigma}$ and the constant map $\Delta_\sigma \rightarrow \Delta_{h\sigma}$ at v_0 .]

(b) Do the same for the normalized standard resolution. [Use the same method as in (a), but make further identifications in X_0 to collapse degenerate simplices. Explicitly, for each $\sigma = (g_0, \dots, g_n)$ let $s_i\sigma = (g_0, \dots, g_i, g_i, \dots, g_n)$ and collapse $\Delta_{s_i\sigma}$ to Δ_σ via the linear map which sends $v_0, \dots, v_{n+1}$ to $v_0, \dots, v_i, v_i, \dots, v_n$.]

6 Periodic Resolutions via Free Actions on Spheres

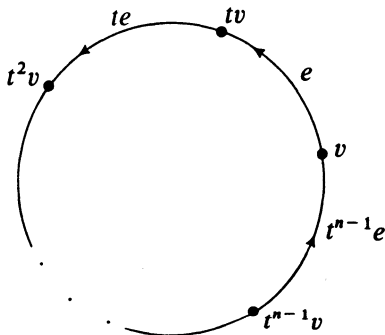
Let X be a free G -complex which is homeomorphic to an odd dimensional sphere S^{2k-1} . (G is then necessarily finite since X is compact.) By an easy special case of the Lefschetz fixed-point theorem (cf. exercise 1 below), G acts trivially on $H_{2k-1}X = \mathbb{Z}$. Writing $C_* = C_*(X)$, we have then an exact sequence of G -modules

$$(6.1) \quad 0 \rightarrow \mathbb{Z} \xrightarrow{\eta} C_{2k-1} \rightarrow \cdots \rightarrow C_1 \rightarrow C_0 \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0,$$

where each C_i is free. Splicing together an infinite number of copies of 6.1, we obtain a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ which is periodic of period $2k$:

$$(6.2) \quad \cdots \rightarrow C_1 \rightarrow C_0 \xrightarrow{\eta\epsilon} C_{2k-1} \rightarrow \cdots \rightarrow C_1 \rightarrow C_0 \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0.$$

EXAMPLE. Let G be a finite cyclic group of order n with generator t . Then G acts freely as a group of rotations of S^1 , regarded as a CW -complex with n vertices and n 1-cells:



Note that $H_1 S^1$ is generated by the cycle $e + te + t^2e + \cdots + t^{n-1}e = Ne$, where N is the "norm element" $1 + t + \cdots + t^{n-1}$ of $\mathbb{Z}G$, so 6.1 takes the form

$$0 \rightarrow \mathbb{Z} \xrightarrow{\eta} \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0,$$

where $\varepsilon(1) = 1$ and $\eta(1) = N$. We therefore obtain the following periodic resolution of period 2, which the reader has probably already seen (§2, exercise 2):

$$(6.3) \quad \cdots \rightarrow \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \xrightarrow{N} \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0.$$

We will see other examples of groups with periodic resolutions in Chapter VI.

EXERCISES

1. Prove the following special case of the Lefschetz fixed-point theorem: Let X be a finite CW -complex and $f: X \rightarrow X$ a map such that, for every cell σ ,

$$f(\sigma) \subseteq \bigcup_{\substack{\tau \neq \sigma \\ \dim \tau \leq \dim \sigma}} \tau.$$

(In particular, f has no fixed-points.) Then the Lefschetz number $\sum (-1)^i \text{trace } f_i$ is zero, where f_i is the endomorphism induced by f on the free abelian group $H_i X / \text{torsion}$. [Hint: The Lefschetz number can be computed on the chain level, where the matrix of f has zeroes on the diagonal.] If $H_*(X) \approx H_*(S^{2n-1})$, deduce that $f_*: H_{2n-1} X \rightarrow H_{2n-1} X$ must be the identity.

2. Prove that the group of order 2 is the only non-trivial group which can act freely on an even-dimensional sphere.

7 Uniqueness of Resolutions

We return to the generality of §1, i.e., we work over an arbitrary ring R . It is obvious that a given R -module M admits many different free resolutions; the purpose of this section is to show that all such resolutions are homotopy equivalent. In the course of proving this we will be faced with mapping problems which can be put in the form

$$(7.1) \quad \begin{array}{ccccc} & & P & & \\ & \swarrow \psi & \downarrow \varphi & \searrow 0 & \\ M' & \xrightarrow{i} & M & \xrightarrow{j} & M'', \end{array}$$

where the solid arrows represent given maps (with $j\varphi = 0$) and the dotted arrow represents a map we would like to construct. A solution to this problem, then, consists of a map $\psi: P \rightarrow M'$ such that $i\psi = \varphi$. A module P is called *projective* if a solution exists for every mapping problem 7.1 in which the row is exact. More concisely, P is projective if the functor $\text{Hom}_R(P, -)$ is exact.

For our present purposes, the main interest in projectivity is provided by:

(7.2) Lemma. *Free modules are projective.*

PROOF. Let F be free with basis (e_a) , and consider a mapping problem

$$\begin{array}{ccccc} & & F & & \\ & \swarrow \psi & \downarrow \varphi & \searrow 0 & \\ M' & \xrightarrow{i} & M & \xrightarrow{j} & M'' \end{array}$$

with exact row. Then $\varphi(e_a) \in \ker j = \text{im } i$, so we can choose $x_a \in M'$ with $i(x_a) = \varphi(e_a)$. Now let ψ be the unique R -module map with $\psi(e_a) = x_a$. $\square$

The next lemma treats the particular mapping problems which one has to solve in trying to construct chain maps and homotopies inductively:

(7.3) Lemma. (a) *Suppose given a diagram*

$$\begin{array}{ccccc} P & \xrightarrow{d} & Q & & \\ \downarrow g & & \downarrow f & & \\ M' & \xrightarrow{d_1} & M & \xrightarrow{d_2} & M'', \end{array}$$

where $d_2 f d = 0$ and it is desired to find a g such that $d_1 g = f d$. If P is projective and the bottom row is exact, then such a g exists.

(b) *Suppose given a diagram (not necessarily commutative)*

$$\begin{array}{ccccc} P & \xrightarrow{d} & Q & & \\ \swarrow k & \downarrow f & \searrow h & & \\ M' & \xrightarrow{d_1} & M & \xrightarrow{d_2} & M'', \end{array}$$

where $d_2 h d = d_2 f$ and it is desired to find a k such that $d_1 k + h d = f$. If P is projective and the bottom row is exact, then such a k exists.

PROOF. (a) is obvious, since the given mapping problem is of the form 7.1 with $\varphi = f d$. Similarly, (b) is a problem of the form 7.1 with $\varphi = f - h d$. $\square$

We can now prove the “fundamental lemma of homological algebra,” which says, roughly speaking, that it is easy to construct chain maps and homotopies from a projective complex to an acyclic one:

(7.4) Lemma. *Let (C, ∂) and (C', ∂') be chain complexes and let r be an integer. Let $(f_i: C_i \rightarrow C'_i)_{i \leq r}$ be a family of maps such that $\partial'_i f_i = f_{i-1} \partial_i$ for $i \leq r$. If C_i is projective for $i > r$ and $H_i(C') = 0$ for $i \geq r$, then $(f_i)_{i \leq r}$ extends to a chain map $f: C \rightarrow C'$, and f is unique up to homotopy. More precisely, any two extensions are homotopic by a homotopy h such that $h_i = 0$ for $i \leq r$.*

PROOF. Assume inductively that f_i has been defined for $i \leq n$, where $n \geq r$; and that $\partial'_i f_i = f_{i-1} \partial_i$ for $i \leq n$. We then have a mapping problem

$$\begin{array}{ccccc} C_{n+1} & \xrightarrow{\partial} & C_n & \xrightarrow{\partial} & C_{n-1} \\ \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} \\ C'_{n+1} & \xrightarrow{\partial'} & C'_n & \xrightarrow{\partial'} & C'_{n-1}, \end{array}$$

where $\partial' f_n \partial = f_{n-1} \partial \partial = 0$. The desired f_{n+1} therefore exists by 7.3a.

Suppose now that g is a second extension of $(f_i)_{i \leq r}$. We wish to find a homotopy h between f and g . Assume inductively that $h_i: C_i \rightarrow C'_{i+1}$ has been defined for $i \leq n$, where $n \geq r$, and that $\partial' h_i + h_{i-1} \partial = f_i - g_i$. (Note that we can start the induction by setting $h_i = 0$ for $i \leq r$.) Setting $\tau_i = f_i - g_i$, we then have a mapping problem

$$\begin{array}{ccccc} C_{n+1} & \xrightarrow{\partial} & C_n & \xrightarrow{\partial} & C_{n-1} \\ \swarrow h_{n+1} \quad \tau_{n+1} \downarrow & & \swarrow h_n \quad \tau_n \downarrow & & \swarrow h_{n-1} \quad \tau_{n-1} \downarrow \\ C'_{n+2} & \xrightarrow{\partial'} & C'_{n+1} & \xrightarrow{\partial'} & C'_n \end{array}$$

with

$$\begin{aligned} \partial' h_n \partial &= (\tau_n - h_{n-1} \partial) \partial && \text{by the inductive hypothesis} \\ &= \tau_n \partial && \text{since } \partial^2 = 0 \\ &= \partial' \tau_{n+1} && \text{since } \tau \text{ is a chain map.} \end{aligned}$$

The desired h_{n+1} with $\partial' h_{n+1} + h_n \partial = \tau_{n+1}$ therefore exists by 7.3b. $\square$

Remark. The proof of 7.4 should have looked familiar to anyone who has seen the method of acyclic models in algebraic topology. We will explain this “similarity” in exercise 3 below.

Now let $\varepsilon: F \rightarrow M$ and $\varepsilon': F' \rightarrow M$ be two free (or projective) resolutions of a module M . We can then form the augmented chain complexes with M in dimension -1 and apply 7.4 with $r = -1$:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & F_1 & \longrightarrow & F_0 & \xrightarrow{\varepsilon} & M \longrightarrow 0 \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \text{id}_M \\ \cdots & \longrightarrow & F'_1 & \longrightarrow & F'_0 & \xrightarrow{\varepsilon'} & M \longrightarrow 0 \longrightarrow \cdots \end{array}$$

We conclude that there is a chain map $f: F \rightarrow F'$ which is *augmentation-preserving*, i.e., which satisfies $\varepsilon' f = \varepsilon$. Moreover, f is unique up to homotopy. (Note that the homotopy h given by 7.4 on the level of *augmented* chain

complexes yields a homotopy $F \rightarrow F'$, because $h_{-1} = 0$.) Similarly, there is an augmentation-preserving map $f': F' \rightarrow F$, and we have $f'f \simeq \text{id}_F$ and $ff' \simeq \text{id}_{F'}$, again by the uniqueness part of 7.4. This proves:

(7.5) Theorem. *Given projective resolutions F and F' of a module M , there is an augmentation-preserving chain map $f: F \rightarrow F'$, unique up to homotopy, and f is a homotopy equivalence.*

We will express this informally by saying that projective resolutions are “unique up to canonical homotopy equivalence.” At the moment, of course, we are mainly interested in *free* resolutions, but later (beginning in Chapter VIII) we will need to consider projective resolutions which are not known to be free.

For future reference we record two special cases of 7.5:

(7.6) Corollary. *Let $\varepsilon: F \rightarrow \mathbb{Z}$ be a free resolution of $\mathbb{Z}$ as a $\mathbb{Z}$ -module. Then ε is a homotopy equivalence.*

PROOF. $\text{id}_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$ is also a free resolution, and $\varepsilon: F \rightarrow \mathbb{Z}$ can be viewed as an augmentation-preserving chain map from one resolution to another. $\square$

(7.7) Corollary. *If F is a non-negative acyclic chain complex of projective modules (over an arbitrary ring R), then F is contractible.*

PROOF. F and the zero complex are projective resolutions of 0, so they are homotopy equivalent. See also exercise 3 of §8 for an alternative proof. $\square$

Remarks

1. In practice, a free resolution usually comes equipped with an explicit basis in each dimension, and one usually proves that it is a resolution by exhibiting a contracting homotopy for the underlying augmented chain complex of $\mathbb{Z}$ -modules. In this case, the proof of 7.5 yields a specific map $f: F \rightarrow F'$. Namely, if k is a contracting homotopy for the augmented complex associated to F' , then f is determined inductively by $f_{n+1}(e_\alpha) = k_n f_n \partial e_\alpha$, where (e_α) is a basis for F_{n+1} .

2. The uniqueness theorem 7.5 can be thought of as the algebraic analogue of Hurewicz's theorem, which we quoted in the introduction, asserting the uniqueness up to homotopy of an aspherical space with a given fundamental group.

EXERCISES

1. Let G be a finite cyclic group. Let F be the free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ given in 6.3 and let F' be the bar resolution. Write down an augmentation preserving chain map $f: F \rightarrow F'$.

2. The method of proof of 7.4 applies to many situations, some of which will arise later in this book. It is therefore useful to axiomatize 7.4, as follows. By an *additive category* we mean a category $\mathcal{A}$ in which $\text{Hom}(A, B)$ is endowed with an abelian group structure for any two objects A, B , in such a way that the composition law $\text{Hom}(B, C) \times \text{Hom}(A, B) \rightarrow \text{Hom}(A, C)$ is $\mathbb{Z}$ -bilinear. In particular, since we have a zero element $0 \in \text{Hom}(A, B)$ it is clear what we mean by a *chain complex* in $\mathcal{A}$. The usual definitions of *chain map* and *homotopy* also go through with no change. [Note that additivity is needed to define “homotopy.”] Suppose now that we are given a class $\mathcal{E}$ of sequences $M' \rightarrow M \rightarrow M''$ in $\mathcal{A}$ called *exact sequences*. We then say that an object P of $\mathcal{A}$ is *projective* relative to $\mathcal{E}$ if every mapping problem 7.1 whose row is in $\mathcal{E}$ admits a solution. Verify that the proof of 7.4 is valid in this situation and leads to an analogue of 7.4 for chain complexes in $\mathcal{A}$. [Note: In the statement of this analogue, the “acyclicity” hypothesis on C' should be replaced by the assumption that $C'_{i+1} \rightarrow C'_i \rightarrow C'_{i-1}$ is in $\mathcal{E}$ for $i \geq r$.] Deduce an analogue of 7.5.

3. Let $\mathcal{C}$ be an arbitrary category and let $\mathcal{A}$ be the additive category whose objects are covariant functors $\mathcal{C} \rightarrow (\text{abelian groups})$ and whose maps are natural transformations of functors. Fix a subclass $\mathcal{M}$ of the class of objects of $\mathcal{C}$. A sequence $T' \rightarrow T \rightarrow T''$ in $\mathcal{A}$ will be called *$\mathcal{M}$ -exact* if the resulting sequence $T'(M) \rightarrow T(M) \rightarrow T''(M)$ of abelian groups is exact for all $M \in \mathcal{M}$. The purpose of this exercise is to show that when exercise 2 is applied to $\mathcal{A}$ (with $\mathcal{E}$ equal to the class of $\mathcal{M}$ -exact sequences), the result is the acyclic models theorem as given, for instance, in Spanier [1966], Theorem 4.2.8, or Dold [1972], Proposition VI.11.7. The crucial step is to prove an analogue in $\mathcal{A}$ of Lemma 7.2; this is done in (a) and (b) below.

(a) (Yoneda’s lemma) Let A be an object of $\mathcal{C}$ and let $h_A: \mathcal{C} \rightarrow (\text{sets})$ be the covariant functor *represented* by A , i.e., $h_A = \text{Hom}_{\mathcal{C}}(A, -)$. Let $u_A \in h_A(A)$ be the identity map $A \rightarrow A$. Let $T: \mathcal{C} \rightarrow (\text{sets})$ be an arbitrary covariant functor. For any $v \in T(A)$, show that there is a unique natural transformation $\varphi: h_A \rightarrow T$ such that $\varphi(u_A) = v$. Thus $\text{Hom}_{\mathcal{F}}(h_A, T) \approx T(A)$, where $\mathcal{F}$ is the category of functors $\mathcal{C} \rightarrow (\text{sets})$. [This can be thought of as saying that h_A is “freely generated” by u_A . The proof is straightforward definition-checking. To prove uniqueness, for example, note that we must have $\varphi(f) = T(f)(v)$ for any $f \in h_A(B) = \text{Hom}_{\mathcal{C}}(A, B)$, in view of the diagram

$$\begin{array}{ccc} h_A(A) & \xrightarrow{h_A(f)} & h_A(B) \\ \varphi \downarrow & & \downarrow \varphi \\ T(A) & \xrightarrow{T(f)} & T(B). \end{array}$$

To prove existence, take this equation as a definition and check that it works.]

(b) Let $\mathbb{Z}h_A: \mathcal{C} \rightarrow (\text{abelian groups})$ be the composite of h_A with the functor $(\text{sets}) \rightarrow (\text{abelian groups})$ which associates to a set the free abelian group it generates. A functor $F: \mathcal{C} \rightarrow (\text{abelian groups})$ will be called *$\mathcal{M}$ -free* if it is isomorphic to a direct sum $\bigoplus_{\alpha} \mathbb{Z}h_{A_{\alpha}}$, where $A_{\alpha} \in \mathcal{M}$ for all α . Deduce from (a) that $\mathcal{M}$ -free functors are projective relative to the class of $\mathcal{M}$ -exact sequences.

(c) Using (b) and exercise 2, state a theorem about chain maps in $\mathcal{A}$ from $\mathcal{M}$ -free complexes to “ $\mathcal{M}$ -acyclic” complexes. [Note: This theorem is precisely the acyclic models theorem cited above.]

4. Another important example of exercise 2 is obtained by taking $\mathcal{A}$ to be the *dual* of the category of R -modules. Thus $\mathcal{A}$ has one object M° for every R -module M and one map $f^\circ: M_1^\circ \rightarrow M_2^\circ$ for every R -module map $f: M_2 \rightarrow M_1$. Composition is defined by $f^\circ g^\circ = (gf)^\circ$. As exact sequences in $\mathcal{A}$ we take those sequences $M_1^\circ \rightarrow M_2^\circ \rightarrow M_3^\circ$ such that the corresponding sequence $M_3 \rightarrow M_2 \rightarrow M_1$ of R -modules is exact. Applying exercise 2, we get an analogue of 7.4 for chain complexes in $\mathcal{A}$, which can obviously be restated as a theorem about cochain complexes of R -modules. Explicitly state this theorem, as well as the theorem corresponding to 7.5. [Note: An R -module Q is called *injective* if Q° is projective in $\mathcal{A}$, or, equivalently, if the functor $\text{Hom}_R(-, Q)$ is exact. Your theorem should therefore be stated in terms of maps of an acyclic cochain complex into a cochain complex of injectives. To give this theorem substance, of course, we should have an analogue of 7.2, so that we will have examples of injectives. We will provide such an analogue later, in §III.4.]

8 Projective Modules

The reader may be curious at this point to know more about projective modules, other than the fact that free modules are projective. We give in this section, therefore, some miscellaneous results and examples concerning projective modules and complexes. We will not make serious use of these results (except as they apply to free modules) until Chapter VIII. One may therefore skip this section now and return to it later.

The first observation is that to prove a module P is projective one need only consider mapping problems 7.1 in which $M'' = 0$; for 7.1 can be replaced by

$$\begin{array}{ccc} & P & \\ \swarrow \text{dashed} & \downarrow j & \\ M' & \longrightarrow & \ker j \longrightarrow 0. \end{array}$$

Thus we have:

(8.1) Proposition. *P is projective if and only if for every surjection $\pi: M \rightarrow \bar{M}$ and every map $\varphi: P \rightarrow \bar{M}$ there is a map $\psi: P \rightarrow M$ such that $\varphi = \pi\psi$:*

$$\begin{array}{ccc} & P & \\ \swarrow \text{dashed } \psi & \downarrow \varphi & \\ M & \xrightarrow{\pi} & \bar{M}. \end{array}$$

One also has the following characterization of projectivity:

(8.2) Proposition. *The following conditions on an R -module P are equivalent:*

- (i) P is projective.
- (ii) Every exact sequence $0 \rightarrow M' \rightarrow M \rightarrow P \rightarrow 0$ splits.
- (iii) P is a direct summand of a free module.
- (iv) There are elements $e_i \in P$ and $f_i \in \text{Hom}_R(P, R)$ (where i ranges over some index set I) such that for every $x \in P$, $f_i(x) = 0$ for almost all i and $x = \sum_{i \in I} f_i(x)e_i$.

PROOF. If P is projective and we are given an exact sequence as in (ii), then we can split the sequence by lifting $\text{id}: P \rightarrow P$ to a map $P \rightarrow M$. Hence (i) $\Rightarrow$ (ii). Choosing such an exact sequence with M free, we see that (ii) $\Rightarrow$ (iii). It is immediate from the definitions that any direct summand of a projective is projective; so (iii) $\Rightarrow$ (i). Finally, (iv) is simply a restatement of (iii). $\square$

EXAMPLES

1. Let $e \in R$ be *idempotent*, i.e., $e^2 = e$. Then right multiplication by e is a projection operator of R onto the direct summand Re . So the left ideal Re is projective.

2. Let R be a (commutative) integral domain and I an invertible ideal. [This means that there is an R -submodule J of the field of fractions of R such that $IJ = R$, where IJ is the set of finite sums $\sum a_i b_i$, $a_i \in I$, $b_i \in J$.] Then I is a projective module. For if we write $1 = \sum e_i f_i$ where $e_i \in I$, $f_i \in J$, then the criterion of 8.4(iv) is satisfied. [Each f_i gives rise, by multiplication, to a homomorphism $I \rightarrow R$ which plays the role of the f_i in 8.4(iv).] On the other hand, I is free only if it is a principal ideal, since any two elements $a, b \in I$ are linearly dependent.

3. Let $R = \mathbb{Z}[\zeta]$, where ζ is a primitive twenty-third root of unity. It is known from algebraic number theory that R has an ideal I which is not principal and that every non-zero ideal in R is invertible. Hence I is projective but not free.

4. Let G be a cyclic group of prime order p . There is a theorem due to Rim which relates projective modules over $\mathbb{Z}G$ to projective modules over $\mathbb{Z}[\zeta]$, where ζ is a primitive p -th root of unity, cf. Milnor [1971], §3. In particular, if $p = 23$, we deduce from example 3 that $\mathbb{Z}G$ has non-free projectives.

5. If R is the rational group algebra $\mathbb{Q}G$ of a finite group G , then one can show that every R -module is projective, cf. exercise 5 below. As an illustration we will prove that $\mathbb{Q}$, with trivial G -action, is projective. Note first that the functor $\text{Hom}_{\mathbb{Q}G}(\mathbb{Q}, -)$ is simply the “invariants” functor $M \mapsto M^G$, where

$$M^G = \{m \in M : gm = m \text{ for all } g \in G\}.$$

Thus we need to show that any surjection $M \rightarrow \bar{M}$ of $\mathbb{Q}G$ -modules gives rise to a surjection $M^G \rightarrow \bar{M}^G$. This is shown by *averaging*: if $\bar{m} \in \bar{M}^G$, lift $\bar{m}$ to $m \in M$; then $(1/|G|) \sum_{g \in G} gm$ is also a lifting of $\bar{m}$ and is in M^G .

We turn now to the *duality theory* for finitely generated projectives, analogous to that for finite dimensional vector spaces over a field. For any left R -module M , let $M^* = \text{Hom}_R(M, R)$. Here R is regarded as a left R -module in forming Hom , but it also has a right R -module structure which we can use to make M^* a right R -module; namely, we set $(ur)(m) = u(m)r$ for $u \in M^*$, $r \in R$, $m \in M$. Similarly, we can define the dual of a right module, and it is a left module.

The main facts about duality are given in the following proposition. Parts (b) and (c) are the most important ones for our purposes; they allow one to use duality to convert Hom to $\otimes$ and vice-versa.

(8.3) Proposition. *Let P be a finitely generated projective left R -module.*

(a) *P^* is a finitely generated projective right R -module.*

(b) *For any left R -module M , there is an isomorphism*

$$\varphi: P^* \otimes_R M \xrightarrow{\sim} \text{Hom}_R(P, M)$$

of abelian groups, given by $\varphi(u \otimes m)(x) = u(x) \cdot m$ for $u \in P^$, $m \in M$, $x \in P$.*

(c) *For any right R -module M , there is an isomorphism*

$$\varphi': M \otimes_R P \xrightarrow{\sim} \text{Hom}_R(P^*, M),$$

given by $\varphi'(m \otimes x)(u) = m \cdot u(x)$ for $m \in M$, $x \in P$, $u \in P^$.*

(d) *There is an isomorphism*

$$\varphi'': P \xrightarrow{\sim} P^{**}$$

of left R -modules, given by $\varphi''(x)(u) = u(x)$ for $x \in P$, $u \in P^$.*

(In connection with (b) and (c), the reader should recall that one can form $M \otimes_R N$ whenever M is a right module and N is a left module; the tensor products written down above therefore make sense.)

PROOF. It is clear from the proof of 8.2 that P can be written as a direct summand of a finitely generated free module F . By additivity, then, it suffices to prove the proposition for F . In more detail: If $F = P \oplus Q$, then $F^* = P^* \oplus Q^*$, $F^* \otimes_R M = (P^* \otimes_R M) \oplus (Q^* \otimes_R M)$, etc., and the maps φ , φ' , and φ'' preserve these decompositions. So (a)–(d) for P will follow from (a)–(d) for F . [In order to use this argument, of course, one must first check that φ and φ' are well-defined.] By additivity again, it suffices to consider the case where F is free of rank 1, i.e., we may assume $F = R$. In this case $R^* \approx R$, whence (a), and it is easy to verify (b)–(d). To prove (b), for instance, one need only check that φ is the composite of the canonical isomorphisms $R^* \otimes_R M \xrightarrow{\sim} M \xrightarrow{\sim} \text{Hom}_R(R, M)$. $\square$

Next we give some properties of chain complexes of projectives.

(8.4) Theorem. *If $f: P' \rightarrow P$ is a weak equivalence between non-negative complexes of projectives, then f is a homotopy equivalence.*

PROOF. The mapping cone of f is non-negative, projective, and acyclic (cf. 0.6). It is therefore contractible by 7.7, so f is a homotopy equivalence by 0.7. $\square$

Using similar methods, we will prove a closely related mapping property of projective complexes, from which we could have deduced 8.4:

(8.5) Theorem. *Let $f: C' \rightarrow C$ be a weak equivalence between arbitrary complexes. If P is a non-negative complex of projectives, then*

$$\mathcal{H}om_R(P, f): \mathcal{H}om_R(P, C') \rightarrow \mathcal{H}om_R(P, C)$$

is a weak equivalence. In particular, the map $[P, C] \rightarrow [P, C']$ induced by f is an isomorphism.

PROOF. Let C'' be the mapping cone of f . It is acyclic, and the mapping cone of $\mathcal{H}om_R(P, f)$ is $\mathcal{H}om_R(P, C'')$; so it suffices to show that $\mathcal{H}om_R(P, C'')$ is acyclic, i.e., that $[P, C'']_n = 0$ for all $n \in \mathbb{Z}$. Now $[P, C'']_n = [P, \Sigma^{-n}C'']$, and the latter is zero by the uniqueness part of the fundamental lemma 7.4, since any map on P is zero in negative dimensions. $\square$

Finally, we prove an analogue of 8.5 for tensor products. Projectivity is unnecessarily strong for this purpose and can be replaced by a “flatness” hypothesis. Recall that a (left) R -module F is *flat* if the functor $- \otimes_R F$ is exact. Free modules are flat, for example, and hence projectives are flat by 8.2(iii).

(8.6) Theorem. *Let $f: C' \rightarrow C$ be a weak equivalence between complexes of right R -modules. If P is a non-negative complex of flat left R -modules, then $f \otimes_R P: C' \otimes_R P \rightarrow C \otimes_R P$ is a weak equivalence.*

PROOF. Let C'' be the mapping cone of f . It is acyclic, and $C'' \otimes_R P$ is the mapping cone of $f \otimes P$; so it suffices to show that $C'' \otimes_R P$ is acyclic. Let $P^{(n)}$ be the n -skeleton of P , i.e., the truncation $(P_i)_{i \leq n}$. We will show inductively that $C'' \otimes_R P^{(n)}$ is acyclic. Note first that $C'' \otimes_R F$ is acyclic for any complex F consisting of a flat module concentrated in a single dimension, since the exact sequences $C''_{i+1} \rightarrow C''_i \rightarrow C''_{i-1}$ remain exact when tensored with F . But $P^{(n)}/P^{(n-1)}$ is such a complex F . So if we assume inductively that $C'' \otimes_R P^{(n-1)}$ is acyclic, it follows from the exact sequence $0 \rightarrow C'' \otimes_R P^{(n-1)} \rightarrow C'' \otimes_R P^{(n)} \rightarrow C'' \otimes_R (P^{(n)}/P^{(n-1)}) \rightarrow 0$ that $C'' \otimes_R P^{(n)}$ is

acyclic. Finally, $C'' \otimes_R P$ is the increasing union of the acyclic complexes $C'' \otimes_R P^{(n)}$, hence it is acyclic. $\square$

EXERCISES

- For what groups G is $\mathbb{Z}$ a projective $\mathbb{Z}G$ -module? [Hint: When does $\varepsilon: \mathbb{Z}G \rightarrow \mathbb{Z}$ split as a map of G -modules?]
- If P is a projective $\mathbb{Z}G$ -module, show that P is also projective as $\mathbb{Z}H$ -module for any $H \subset G$. [Use criterion 8.2(iii).]
- (a) Use 8.2 to give another proof of 7.7. [According to 0.3, it suffices to show that the surjection $\bar{\partial}_n: P_n \rightarrow Z_{n-1}$ induced by ∂_n splits for all n . Assuming inductively that $\bar{\partial}_{n-1}$ splits, $Z_{n-1} = \ker \bar{\partial}_{n-1}$ is a direct summand of P_{n-1} , hence it is projective. Therefore $\bar{\partial}_n$ splits.]
 (b) Use the same method to show that the non-negativity hypothesis in 7.7 can be dropped for certain rings R , e.g., for principal ideal domains. [If R is a principal ideal domain, then submodules of a projective module are projective (in fact, free). So Z_{n-1} above is automatically projective and we do not need to use induction.]
- If G is a group and X is a G -set such that all isotropy groups G_x are finite, show that the permutation module $\mathbb{Q}X$ is a projective $\mathbb{Q}G$ -module.
- If G is finite and k is a field of characteristic zero, show that every kG -module is projective. [Given an exact sequence as in 8.2(ii), choose a splitting $f: P \rightarrow M$ for the underlying sequence of k -vector spaces. Then $x \mapsto (1/|G|) \sum_{g \in G} gf(g^{-1}x)$ is a kG -splitting.]
- Prove the following converse of 8.3b: If P is a module such that $\varphi: P^* \otimes_R P \rightarrow \text{Hom}_R(P, P)$ is surjective, then P is finitely generated and projective. [Write $\text{id}_P = \varphi(\sum f_i \otimes e_i)$ and show that 8.2(iv) is satisfied.]
- Let P be finitely generated and projective. For any $z \in P^* \otimes_R P$ and any module M there is a map $\psi_z: \text{Hom}_R(P, M) \rightarrow P^* \otimes_R M$, defined as follows: $\psi_z(f)$ is the image of z under $P^* \otimes f: P^* \otimes_R P \rightarrow P^* \otimes_R M$. Show that the inverse of the canonical isomorphism $\varphi: P^* \otimes_R M \xrightarrow{\sim} \text{Hom}_R(P, M)$ of 8.3b is a map of the form ψ_z for some fixed z (independent of M). [Method 1: View φ^{-1} as a natural transformation $\text{Hom}_R(P, -) \rightarrow P^* \otimes_R -$. By Yoneda's lemma (exercise 3a of §7), φ^{-1} is uniquely determined by what it does to id_P . Moreover, the proof of Yoneda's lemma tells you how to describe φ^{-1} in terms of $z = \varphi^{-1}(\text{id}_P)$, and this description says precisely that $\varphi^{-1} = \psi_z$. Method 2: Choose (e_i) and (f_i) as in 8.2(iv) and set $z = \sum f_i \otimes e_i$. By directly checking definitions, verify that $\psi_z \circ \varphi = \text{id}$ and/or that $\varphi \circ \psi_z = \text{id}$.]
- If P is finitely presented and flat, show that P is projective. [Take a finite presentation $F_1 \rightarrow F_0 \rightarrow P \rightarrow 0$ with F_0 and F_1 free of finite rank. This gives an exact sequence $0 \rightarrow P^* \rightarrow F_0^* \rightarrow F_1^*$ of right R -modules. Tensor with P and deduce that $P^* \otimes_R P \xrightarrow{\sim} \text{Hom}_R(P, P)$. Now apply exercise 6.]

Appendix. Review of Regular Coverings

The material summarized in this appendix can be found in many algebraic topology texts, such as Massey [1967] or Spanier [1966].

Let $p: \tilde{X} \rightarrow X$ be a covering map of connected, locally path-connected spaces. A *deck transformation* of p is a homeomorphism $g: \tilde{X} \rightarrow \tilde{X}$ such that $pg = p$. The group G of all deck transformations acts *freely* on $\tilde{X}$, in the sense that $g\tilde{x} \neq \tilde{x}$ for all $\tilde{x} \in \tilde{X}$ and $g \neq 1$ in G .

The cover p is said to be *regular* if it satisfies the following conditions, which are equivalent:

- (i) G acts transitively on $p^{-1}x$ for all $x \in X$. [Hence $X \approx \tilde{X}/G$.]
- (ii) The image of $\pi_1 \tilde{X} \rightarrow \pi_1 X$ is normal in $\pi_1 X$ for some (and hence every) choice of basepoints.
- (iii) For any closed loop ω in X , if one lift of ω to $\tilde{X}$ is closed, then all lifts of ω are closed.

In this case we have $G \approx \pi_1 X / \pi_1 \tilde{X}$. In particular, if $\tilde{X}$ is simply connected (so that p is the *universal cover* of X), then $G \approx \pi_1 X$.

To get an explicit isomorphism $G \approx \pi_1 X / \pi_1 \tilde{X}$ above, we must choose basepoints $x \in X$ and $\tilde{x} \in p^{-1}x$. We then have a homomorphism $\varphi: \pi_1(X, x) \rightarrow G$, defined as follows: Let $\omega: [0, 1] \rightarrow X$ represent $[\omega] \in \pi_1(X, x)$ and let $\tilde{\omega}: [0, 1] \rightarrow \tilde{X}$ be the lift of ω with $\tilde{\omega}(0) = \tilde{x}$. Then $\tilde{\omega}(1) \in p^{-1}x$, and $\varphi([\omega])$ is defined to be the unique element of G such that

$$(A1) \quad \varphi([\omega])\tilde{x} = \tilde{\omega}(1).$$

One verifies that φ is a homomorphism, is surjective, and has kernel $\pi_1 \tilde{X}$, so it induces the required isomorphism.

Finally, we mention a slightly different point of view which is sometimes useful. [For our purposes this will be needed only in exercise 2 of §II.7.] Fix an abstract group G and a connected, locally path-connected space X . By a *regular G -cover* of X (also called a *principal G -bundle* over X) we mean a covering map $p: \tilde{X} \rightarrow X$, where $\tilde{X}$ is not necessarily connected, together with a free G -action on $\tilde{X}$ satisfying condition (i) above.

In case $\tilde{X}$ is connected, such a p is simply a regular cover in the usual sense, together with an isomorphism of G with the group of deck transformations.

We will assume that a basepoint $x \in X$ has been chosen and that all covering spaces $\tilde{X}$ come equipped with a basepoint $\tilde{x} \in p^{-1}x$. We can then define a homomorphism $\varphi: \pi_1 X \rightarrow G$ exactly as in A1 above, the only difference being that φ will not be surjective if $\tilde{X}$ is disconnected. In fact, one checks easily that $G/\text{im } \varphi \approx \pi_0 \tilde{X}$ (isomorphism of G -sets).

The main theorem on regular G -covers (with basepoint) says that they are completely classified by $\varphi \in \text{Hom}(\pi_1 X, G)$:

(A2) Theorem. *Let $\mathcal{C}_G(X)$ be the set of isomorphism classes of pointed, regular G -covers of X . The assignment of φ to p gives a bijection*

$$\mathcal{C}_G(X) \approx \text{Hom}(\pi_1 X, G).$$

(Note: Isomorphisms are required to preserve basepoints, commute with the G action, and commute with the projection onto X .)

SKETCH OF PROOF. Using the usual classification of connected covering spaces in terms of subgroups of $\pi_1 X$, one easily sees that connected, pointed, regular G -covers correspond to surjections $\varphi: \pi_1 X \rightarrow G$. The study of disconnected covers is easily reduced to the connected case by considering the connected components of $\tilde{X}$. $\square$

CHAPTER II

The Homology of a Group

1 Generalities

In homological algebra one constructs homological invariants of algebraic objects by the following process, or some variant of it:

Let R be a ring and T a covariant *additive* functor from R -modules to abelian groups. Thus the map $\text{Hom}_R(M, N) \rightarrow \text{Hom}_Z(TM, TN)$ defined by T is a homomorphism of abelian groups for all R -modules M, N . For any R -module M , choose a free (or projective) resolution $\varepsilon: F \rightarrow M$ and consider the chain complex TF of abelian groups obtained by applying T to F termwise. Now T , being additive, preserves chain homotopies; so we can apply the uniqueness theorem for resolutions (I.7.5) to deduce that the complex TF is independent, up to canonical homotopy equivalence, of the choice of resolution. Passing to homology, we obtain groups $H_n(TF)$ which depend only on T and M (up to canonical isomorphism).

This construction is of no interest, of course, if T is an *exact* functor; for then the augmented complex

$$\cdots \rightarrow TF_1 \rightarrow TF_0 \rightarrow TM \rightarrow 0$$

is acyclic, so that $H_n(TF) = 0$ for $n > 0$ and $H_0(TF) = TM$. Thus we can regard the groups $H_n(TF)$ in the general case as a measure of the failure of T to be exact.

In this chapter we will apply this construction with $R = \mathbb{Z}G$, $M = \mathbb{Z}$, and T equal to the “co-invariants” functor which we will describe in §2 below. This particular choice of R , M , and T is not arbitrary, as we will see, but rather it is a reflection of the topology which motivates the homology theory of groups.

2 Co-invariants

If G is a group and M is a G -module, then the group of *co-invariants* of M , denoted M_G , is defined to be the quotient of M by the additive subgroup generated by the elements of the form $gm - m$ ($g \in G, m \in M$). Thus M_G is obtained from M by “dividing out” by the G -action. (The name “co-invariants” comes from the fact that M_G is the largest *quotient* of M on which G acts trivially, whereas M^G , the group of invariants, is the largest *submodule* of M on which G acts trivially.) In view of exercise 1a of §I.2, we can also describe M_G as M/IM , where I is the augmentation ideal of $\mathbb{Z}G$ and IM denotes the set of all finite sums $\sum a_i b_i$ ($a_i \in I, b_i \in M$).

Still another description of M_G is given by:

$$(2.1) \quad M_G \approx \mathbb{Z} \otimes_{\mathbb{Z}G} M.$$

Here, in order for the tensor product to make sense, we regard $\mathbb{Z}$ as a *right* $\mathbb{Z}G$ -module (with trivial G -action). To prove 2.1, note that in $\mathbb{Z} \otimes_{\mathbb{Z}G} M$ we have the identity $1 \otimes gm = 1 \cdot g \otimes m = 1 \otimes m$; hence there is a map $M_G \rightarrow \mathbb{Z} \otimes_{\mathbb{Z}G} M$ given by $\bar{m} \mapsto 1 \otimes m$, where $\bar{m}$ denotes the image in M_G of an element $m \in M$. On the other hand, using the universal property of the tensor product, we can define a map $\mathbb{Z} \otimes_{\mathbb{Z}G} M \rightarrow M_G$ by $a \otimes m \mapsto a\bar{m}$. These two maps are inverses of one another.

In view of 2.1 and standard properties of the tensor product, we immediately obtain the following two properties of the co-invariants functor:

(2.2) **Right-exactness:** Given an exact sequence $M' \rightarrow M \rightarrow M'' \rightarrow 0$ of G -modules, the induced sequence $M'_G \rightarrow M_G \rightarrow M''_G \rightarrow 0$ is exact.

(2.3) If F is a free $\mathbb{Z}G$ -module with basis (e_i) , then F_G is a free $\mathbb{Z}$ -module with basis $(\bar{e}_i)$.

Finally, we note that the co-invariants functor arises naturally in the topological setting of §I.4:

(2.4) Proposition. *Let X be a free G -complex and let Y be the orbit complex X/G . Then $C_*(Y) \approx C_*(X)_G$.*

PROOF. The projection $C_*(X) \rightarrow C_*(Y)$ induces by passage to the quotient a map $\varphi: C_*(X)_G \rightarrow C_*(Y)$. Now $C_*(X)_G$ has (by 2.3 and the observations in §I.4) a $\mathbb{Z}$ -basis with one basis element for each G -orbit of cells of X . But $C_*(Y)$ also has a $\mathbb{Z}$ -basis with one element for each G -orbit of cells of X , and it is clear that φ maps a basis element of $C_*(X)_G$ to the corresponding basis element of $C_*(Y)$, hence φ is an isomorphism. $\square$

EXERCISES

1. If S is an arbitrary G -set, show that $(\mathbb{Z}S)_G \approx \mathbb{Z}[S/G]$.
2. The freeness hypothesis in 2.4 is unnecessarily strong. Find a weaker hypothesis under which the conclusion remains true. [Hint: Use exercise 1.]

3. Let H be a normal subgroup of G and let M be a G -module.

(a) Show that the action of G on M induces an action of G/H on M_H .

(b) Show that $M_G \approx (M_H)_{G/H}$.

(c) Show that $M_H \approx \mathbb{Z}[G/H] \otimes_{\mathbb{Z}G} M$ as G/H -modules. (Here the *right*-translation action of G on $\mathbb{Z}[G/H]$ is used to form the tensor product, and the *left*-translation action of G/H on $\mathbb{Z}[G/H]$ is used to give the tensor product a G/H -module structure.)

3 The Definition of H_*G

Let G be a group and $\varepsilon: F \rightarrow \mathbb{Z}$ a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. We define the *homology groups* of G by

$$H_i G = H_i(F_G).$$

As we explained in §1, the right-hand side is independent of the choice of resolution, up to canonical isomorphism. For example, suppose G is a finite cyclic group of order n . Using the resolution

$$\cdots \xrightarrow{t-1} \mathbb{Z}G \xrightarrow{N} \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \rightarrow \mathbb{Z} \rightarrow 0$$

of I.6.3, we obtain for F_G the complex

$$\cdots \xrightarrow{0} \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{0} \mathbb{Z}.$$

Thus

$$(3.1) \quad H_i G \approx \begin{cases} \mathbb{Z} & i = 0 \\ \mathbb{Z}_n & i \text{ odd} \\ 0 & i \text{ even, } i > 0. \end{cases}$$

The reader is invited to similarly compute the homology of a free group (cf. I.4.3) or a free abelian group of rank 2 (cf. §I.4, exercise 3). We will treat these examples topologically in the next section.

For any group G we can always take F to be the standard resolution (§I.5), in which case we write $C_*(G)$ for the chain complex F_G . Using 2.3 and the formula I.5.1, we can describe $C_*(G)$ explicitly, as follows: Define an equivalence relation on the $(n+1)$ -tuples $(g_0, \dots, g_n)$ ($g_i \in G$) by setting $(g_0, \dots, g_n) \sim (gg_0, \dots, gg_n)$ for all $g \in G$, and let $[g_0, \dots, g_n]$ denote the equivalence class of $(g_0, \dots, g_n)$. Then $C_n(G)$ has a $\mathbb{Z}$ -basis consisting of the equivalence classes $[g_0, \dots, g_n]$, and $\partial: C_n(G) \rightarrow C_{n-1}(G)$ is given by $\partial = \sum_{i=0}^n (-1)^i d_i$, where

$$d_i[g_0, \dots, g_n] = [g_0, \dots, \hat{g}_i, \dots, g_n].$$

$C_*(G)$, when described in this way, is often called the *homogeneous* chain complex of G , because of the analogy with homogeneous coordinates for projective space.

The *non-homogeneous* description of $C_*(G)$ is obtained by using the bar notation and the formula I.5.2. From this point of view $C_n(G)$ has a $\mathbb{Z}$ -basis consisting of n -tuples $[g_1 | \cdots | g_n]$, and $\partial = \sum_{i=0}^n (-1)^i d_i$, where

$$d_i[g_1 | \cdots | g_n] = \begin{cases} [g_2 | \cdots | g_n] & i = 0 \\ [g_1 | \cdots | g_i g_{i+1} | \cdots | g_n] & 0 < i < n \\ [g_1 | \cdots | g_{n-1}] & i = n. \end{cases}$$

Note that the symbol $[g_1 | \cdots | g_n]$, which previously was used to denote a typical $\mathbb{Z}G$ -basis element of F_n , now denotes the image of that basis element in $(F_n)_G = C_n(G)$. This abuse of notation, which is standard, might occasionally cause confusion. The reader is also warned that some authors write $[g_1, \dots, g_n]$ or $(g_1, \dots, g_n)$ instead of $[g_1 | \cdots | g_n]$.

In low dimensions $C_*(G)$ has the form

$$C_2(G) \xrightarrow{\partial} C_1(G) \xrightarrow{\partial} \mathbb{Z},$$

where $\partial[g|h] = [h] - [gh] + [g]$. Consequently, $H_0 G = \mathbb{Z}$ and $H_1 G$ is isomorphic to the *abelianization* $G_{ab} = G/[G, G]$. (Explicitly, if we denote by $\bar{g}$ the homology class of the cycle $[g]$, the reader can easily check that there is an isomorphism $H_1 G \rightarrow G_{ab}$ such that $\bar{g} \mapsto g \bmod [G, G]$.)

EXERCISES

1. Let $g_1, \dots, g_n$ be n elements of G which pairwise commute, and let

$$z = \sum (-1)^{\text{sgn } \sigma} [g_{\sigma(1)} | \cdots | g_{\sigma(n)}]$$

in $C_n(G)$, where σ ranges over all permutations of $\{1, \dots, n\}$. (If $n = 2$, for example, $z = [g_1 | g_2] - [g_2 | g_1]$.) Verify that z is a cycle. [Such cycles play an important role in the homology theory of abelian groups, as we will see in Chapter V.]

2. If G is a non-trivial finite cyclic group, show that $\mathbb{Z}$ does not admit a projective resolution of finite length over $\mathbb{Z}G$.
3. If G has torsion (i.e., non-trivial elements of finite order), show that $\mathbb{Z}$ does not admit a projective resolution of finite length over $\mathbb{Z}G$. [Hint: This follows from exercise 2.]

4 Topological Interpretation

If Y is a $K(G, 1)$ -complex with universal cover X , then we know (I.4.2) that $C_*(X)$ is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. Since $C_*(X)_G \approx C_*(Y)$ by 2.4 above, we obtain:

(4.1) Proposition. *If Y is a $K(G, 1)$ -complex then $H_* G \approx H_* Y$.*

[In some treatments of the homology theory of groups, this result is taken as the *definition* of $H_* G$, as we indicated in the introduction.]

EXAMPLES

1. Let Y be a bouquet of circles, indexed by a set S . Then, as we saw in I.4.3, Y is a $K(F(S), 1)$. Hence

$$H_i(F(S)) = H_i Y = \begin{cases} \mathbb{Z} & i = 0 \\ \mathbb{Z}S (= F(S)_{ab}) & i = 1 \\ 0 & i > 1 \end{cases}$$

2. Let g be an integer ≥ 1 and let $G = \langle a_1, \dots, a_g, b_1, \dots, b_g; \prod_{i=1}^g [a_i, b_i] \rangle$, i.e., G is the group with generators $a_1, \dots, a_g, b_1, \dots, b_g$ and the single defining relation $\prod_{i=1}^g [a_i, b_i] = 1$. Thus G is the fundamental group of the closed orientable surface Y of genus g (cf. Massey [1967], ch. 4, §5), and I claim that Y is a $K(G, 1)$. To see this, we need only note that the universal cover X of Y is a *non-compact* surface, since G is infinite. Consequently, one knows from the homology theory of manifolds (cf. Dold [1972], ch. VIII, §3) that $H_i X = 0$ for $i \geq 2$, so Y is a $K(G, 1)$. [Alternatively, one can explicitly exhibit X as the hyperbolic plane tiled by $4g$ -sided polygons (cf. Siegel [1971], ch. 3, §9), hence X is contractible.] Thus

$$H_i G = H_i Y = \begin{cases} \mathbb{Z} & i = 0, 2 \\ \mathbb{Z}^{2g} & i = 1 \\ 0 & i > 2. \end{cases}$$

The interested reader can similarly treat the group $\langle c_1, \dots, c_k; \prod_{i=1}^k c_i^2 \rangle$ ($k \geq 2$), which is the fundamental group of the non-orientable closed surface with k crosscaps. Finally, we remark that non-compact surfaces and surfaces with boundary are also easily seen to be $K(G, 1)$'s, but we obtain no new examples in this way since the fundamental groups are free.

3. The surface groups just considered are examples of *one-relator* groups. Suppose now that $G = \langle S; r \rangle$ is an arbitrary one-relator group, i.e., G is the quotient of a free group $F(S)$ by the normal closure of a single element r . Let Y be the 2-complex $(\bigvee_{s \in S} S_s^1) \cup_r e^2$ obtained from the bouquet of circles $\bigvee S_s^1$ by attaching a 2-cell via the map $S^1 \rightarrow \bigvee S_s^1$ corresponding to r . Then $\pi_1 Y = G$ (cf. Massey [1967], ch. 7, §2). If r is not a power u^n ($n > 1$) in $F(S)$, then a deep theorem of Lyndon implies that Y is a $K(G, 1)$; proofs can be found in Lyndon [1950], Dyer-Vasquez [1973], and Lyndon-Schupp [1977] (ch. III, §§9–11). (If r is a power, on the other hand, then one can show that G has torsion, so that there cannot exist a finite-dimensional $K(G, 1)$, cf. exercise 3 of §3 above.) The chain complex $C_* Y$ is easily seen to have the form

$$\mathbb{Z} \xrightarrow{\partial} \mathbb{Z}S \xrightarrow{\partial} \mathbb{Z},$$

where $\partial(1)$ is the image of r in $\mathbb{Z}S = F(S)_{ab}$. Hence $H_0G = \mathbb{Z}$, $H_1G = G_{ab}$,

$$H_2G = \begin{cases} \mathbb{Z} & \text{if } r \in [F(S), F(S)] \\ 0 & \text{otherwise,} \end{cases}$$

and $H_iG = 0$ for $i > 2$.

4. If $G = \mathbb{Z}^n$, the free abelian group of rank n , then the n -dimensional torus $Y = S^1 \times \cdots \times S^1$ (n factors) is a $K(G, 1)$ since its universal cover $\mathbb{R}^n$ is contractible. Hence H_iG is a free abelian group of rank $\binom{n}{i} = n!/i!(n-i)!$.

Example 4 can be described in terms of the embedding of the group $G = \mathbb{Z}^n$ as a discrete subgroup of the Lie group $L = \mathbb{R}^n$. Indeed, the $K(G, 1)$ $Y = S^1 \times \cdots \times S^1$ is simply the quotient L/G . Our remaining examples will further illustrate this method of constructing $K(G, 1)$'s. These examples will require somewhat more effort to read than the previous ones, and they will not be referred to again until Chapter VIII; the reader may therefore want to simply glance at them now and read them more carefully later.

5. Let G be the $n \times n$ strict upper triangular group over $\mathbb{Z}$, i.e., G is the group of $n \times n$ integral matrices with 1's on the diagonal and 0's below the diagonal. Let L be the $n \times n$ strict upper triangular group over $\mathbb{R}$. Then G is a discrete subgroup of L , and we can form the coset space L/G . Since G is discrete, the projection $L \rightarrow L/G$ is a covering map. Indeed, let U be a neighborhood of 1 in L such that $U \cap G = \{1\}$, and let V be a neighborhood of 1 such that $V^{-1}V \subseteq U$. Then for any $l \in L$, the neighborhood $W = lV$ of l has the property that its transforms Wg ($g \in G$) are disjoint; our assertion follows at once. Finally, L is obviously homeomorphic to Euclidean space $\mathbb{R}^d$, $d = n(n-1)/2$, so the manifold L/G is a $K(G, 1)$ and $H_*G \approx H_*(L/G)$. [Strictly speaking, we should verify that L/G admits a CW -structure, since 4.1 was proved only for $K(G, 1)$ -complexes. This is in fact true by Whitehead's triangulation theorem for smooth manifolds (cf. Munkres [1966], ch. II), but an easier way to deal with the problem is to simply observe that the proof of 4.1 goes through with no difficulty in the context of *singular* homology theory; see exercise 1 below.]

6. Let L be the Lie group $GL_n(\mathbb{R})$. In contrast to the Lie groups considered in examples 4 and 5, L is *not* contractible. Nevertheless, there is a contractible manifold X associated to L which can be used to study the homology of discrete subgroups of L . In order to describe X , we need to recall some elementary linear algebra.

Recall that a *quadratic form* on $\mathbb{R}^n$ is a function $Q: \mathbb{R}^n \rightarrow \mathbb{R}$ of the form

$$Q(x_1, \dots, x_n) = \sum_{i,j=1}^n a_{ij}x_ix_j.$$

The matrix $A = (a_{ij})$ can be taken to be symmetric, and it is then uniquely determined by Q . We will often identify Q with A . The form Q and the matrix A are called *positive definite* if $Q(x) > 0$ for all $x \neq 0$ in $\mathbb{R}^n$. The set of positive

definite symmetric matrices is a convex open subset of the space of all symmetric $n \times n$ matrices.

We now define X to be the space of positive definite quadratic forms on $\mathbb{R}^n$, topologized as a subspace of the space of symmetric $n \times n$ matrices. It is clear from what we have said above that X is a contractible manifold of dimension $d = n(n + 1)/2$. [In fact, $X \approx \mathbb{R}^d$.] There is a right action of $L = GL_n(\mathbb{R})$ on $\mathbb{R}^n$ by right matrix multiplication (where an element of $\mathbb{R}^n$ is thought of as a row vector), and this induces a left action of L on X , said to be given by “change of variable”:

$$(gQ)(x) = Q(xg)$$

for $Q \in X, g \in L, x \in \mathbb{R}^n$. [In terms of symmetric matrices, this action takes the form

$$g \cdot A = gAg^t,$$

where g^t is the transpose of g .] It is well-known that any $Q \in X$ is equivalent under change of variable to the standard form $Q_0 = \sum x_i^2$, so the action of L on X is transitive. Moreover, the isotropy group L_{Q_0} is the orthogonal group $K = O_n(\mathbb{R})$. We therefore have a bijection

$$X \approx L/K$$

of left L -spaces, which can be shown to be a homeomorphism.

Since K is compact, it follows that the action of L on X is *proper*, i.e., that the following condition is satisfied: For every compact set $C \subseteq X$, $\{g \in L: gC \cap C \neq \emptyset\}$ is a compact subset of L . [This says, roughly speaking, that the transforms gC of any compact set C tend to ∞ in X as $g \rightarrow \infty$ in L .]

Suppose now that G is a *discrete* subgroup of L . Then for any compact $C \subseteq X$, $\{g \in G: gC \cap C \neq \emptyset\}$ is finite. One deduces easily that the isotropy group G_x of any $x \in X$ is finite and that x has a neighborhood U such that $gU \cap U = \emptyset$ for $g \in G - G_x$.

Finally, suppose further that G is *torsion-free*. Then the finite isotropy groups G_x must be trivial, and it follows at once that the projection $X \rightarrow X/G$ is a regular covering map with group G . Thus X/G is a $K(G, 1)$ and $H_*G \approx H_*(X/G)$.

This discussion does not apply to $GL_n(\mathbb{Z})$, the most obvious discrete subgroup of $GL_n(\mathbb{R})$, because it is not torsion-free. But $GL_n(\mathbb{Z})$ does have torsion-free subgroups G of finite index (cf. exercise 3 below), and our discussion applies to them. Unfortunately, it is extremely difficult in practice to actually compute $H_*(X/G)$.

7. Consider now the special linear group $SL_n(\mathbb{R}) = \{g \in GL_n(\mathbb{R}): \det g = 1\}$. We can then replace the space X of example 6 by a contractible manifold X_0 on which $SL_n(\mathbb{R})$ acts properly, with $\dim X_0 = \dim X - 1$. Namely, we take X_0 to be the quotient space of X obtained by identifying two quadratic forms which are (positive) scalar multiples of one another. One can verify that

$$X_0 \approx SL_n(\mathbb{R})/SO_n(\mathbb{R}),$$